



UnionPay Integrated Circuit Card Specifications — Auxiliary Specifications

Part I Personalization Template Based on Debit/Credit Application

Version 2017

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Summary of Revisions

The changes listed below are associated with the current version.

Change Description	Revised/ Deleted/ Added	Location
Revised the value of CVM List (tag '8E') to support signature.	Revised	4.1.1, 4.1.12
Added Template 10 for UnionPay debit cards and pre-paid cards issued in the U.S. with the U.S. Common Debit AID added.	Added	4.10
Added the definition of PAR (tag '9F24') in Templates 1-7, 9-10.	Added	4.1.1, 4.2.1, 4.3.1, 4.4.1, 4.5.1, 4.6.1, 4.7.1, 4.9.1
Clarified the value of the Applications Label (tag '50') for UnionPay ICCs and other mobile payment products according to the credit/debit attribute of a specific product in Templates 1-7, 9.	Revised	4.1.10, 4.2.10, 4.3.10, 4.4.10, 4.5.10, 4.6.10, 4.7.10, 4.9.10
Enabled Byte 2 Bit 8 of Card Transaction Qualifiers (tag '9F6C') to indicate whether CDCVM is performed in Templates 11-17.	Revised	5.1.3.5, 5.2.2.5, 5.3.2.5, 5.4.2.5, 5.5.2.5, 5.6.2.5, 5.7.1.5
For Card Additional Processes (tag '9F68') in Templates 11-17: Enabled its Byte 4 Bit 6 to indicate whether ODA for online authorization is supported in qUICS process; Disabled its Byte 1 Bit 6 and set this bit to RFU; Revised its overall setting for PIN-based cards and signature-based cards.	Revised	5.1.3.1, 5.1.3.4, 5.2.2.1, 5.2.2.4, 5.3.2.1, 5.3.2.4, 5.4.2.1, 5.4.2.4, 5.5.2.1, 5.5.2.4, 5.6.2.1, 5.6.2.4, 5.7.1.1, 5.7.1.4
For Card Additional Function Indicator (tag 'DF61') in Templates 12-17: Enabled its Byte 1 Bit 7 to indicate whether ODA for online authorization is supported in qUICS process; Enabled its Byte 1 Bit 5 for the contactless credit-mode low-value payment to indicate whether the card supports credit-mode low-value payments.	Revised	5.2.2.7, 5.3.2.7, 5.4.2.7, 5.5.2.7, 5.6.2.7, 5.7.1.7
Enabled Byte 2 Bit 2 of Application Default Action (tag '9F52') to indicate whether the tag '9F79'/'DF79' will be reset automatically in GENERATE AC processing in Templates 1-7, 9.	Revised	4.1.13, 4.2.14, 4.3.14, 4.4.14, 4.5.14, 4.6.14, 4.7.14, 4.9.13
Adjusted IAC-online and IAC-default in low-value payment based debit/credit templates. Set Byte 1 Bit 3 of IAC-online and IAC-default to 1 to ensure the transaction is transmitted for online processing in case of CDA failure in Templates 11-16.	Revised	5.1.2.5, 5.2.1.5, 5.3.1.5, 5.4.1.5, 5.5.1.5, 5.6.1.5

1 Application Scope

This book applies to all UPI Participants.

2 References

The terms and conditions of the following documents quoted by this document have become the terms and conditions of the document. The modification list (excluding corrected contents) or revised edition attached to the dated documents shall not apply to this Document. However, Participants may study whether to apply the latest versions of such documents. The latest versions of non-dated documents shall apply to the document.

UnionPay Integrated Circuit Card Specifications – Basic Specifications

UnionPay Integrated Circuit Cards Specifications – Product Specifications Part II& Part VI

3 Notations and Abbreviations

3.1 Symbols

3.1.1 Hexadecimal Notation

All data represented in hexadecimal form are enclosed in single quotes. For example: decimal number 27509 converting to hexadecimal is '6B75'.

3.1.2 Binary Notation

A small letter "b" follows all values represented in binary. For example: hexadecimal '08' represented in binary is 00001000b.

3.1.3 Compressed Numeric Data

If compressed number is with an odd number of digits, a hexadecimal 'F' should be filled at the end. For example decimal 456 represented in the compressed data type is '456F', stored in two bytes.

3.1.4 Data Notation

Formats to be supported include:

- n (numeric)
- cn (compressed numeric)
- b (binary)
- an (alphanumeric)
- ans (special alphanumeric)
- var. (variable)

When the length defined for data exceeds that of actual data and when digits are not fully filled, rules of filling gaps are as follows:

- Data element with format n aligned to the right, fill 0 to the left;
- Data element with format cn aligned to the left, fill F to the right;
- Data element with format an aligned to the left, fill 0 to the right;
- Data element with format ans aligned to the left, fill 0 to the right.

3.2 Abbreviations

The following abbreviations are applicable in this document.

AAC	Application Authentication Cryptogram
ADA	Application Default Action
ADF	Application Definition File

AFL	Application File Locator
AID	Application Identifier
ARQC	Authorization Request Cryptogram
ATC	Application Transaction Counter
AUC	Application Usage Control
CAM	Card Authentication Method
CDA	Combined DDA/Application Cryptogram Generation, a form of offline dynamic data authentication.
CDCVM	Consumer Device Cardholder Verification Method
CVM	Cardholder Verification Method
CVN	Cryptogram Version No.
CDOL	Card Risk Management Data Object List
CPLC	Card Production Life Cycle History File Identifiers
CTAL	Cumulative Transaction Amount Limit
DDA	Dynamic Data Authentication, a form of offline data authentication.
DDOL	Dynamic Data Authentication Data Object List
EMV	Europay, MasterCard, Visa
GAC	GENERATE AC
Hex	Hexadecimal
IAC	Issuer Action Code
ICC	Integrated Circuit Card
IPK	Issuer Public Key
LCOL	Lower Consecutive Offline Limit
MDK	Master Derivation Key

ODA	Offline Data Authentication
PAN	Primary Account Number
PAR	Payment Account Reference
PIN	Personal Identification Number
PIX	Proprietary Identifier Extension
PSE	Payment System Environment
qUICS	Contactless Quick Payment Process
RID	Registered Application Provider Identifier
RFU	Reserve for Future Use
SDA	Static Data Authentication, a form of offline data authentication.
SAD	Signed Static Application Data
TC	Transaction Certificate
UCOL	Upper Consecutive Offline Limit
UDK	Unique UICS card level key generated by MDK
UICS	<i>UnionPay Integrated Circuit Card Specifications</i>

4 Standard UICS Debit/Credit Application Personalization Templates

To assist UPI Members in setting card parameters during card issuance, UPI has compiled these templates as a guide to demonstrate the technical settings.

Templates described in this document are widely used by UPI Members. There are detailed descriptions of personalization in *Debit/Credit Application Personalization Guide*.

This document has defined minimal data set indispensable to each template, and each data object requires a data value, which is provided by the template, based on the best recommendations or must be specified by the issuer.

Template 1	For debit cards, DDA, online PIN+signature, bypass PIN allowed
Template 2	For debit cards, DDA, online PIN, bypass PIN not allowed
Template 3	For credit cards, DDA, online PIN, bypass PIN allowed
Template 4	For credit cards, DDA, signature, online control
Template 5	For credit cards, DDA, online PIN, bypass PIN not allowed
Template 6	For credit cards, DDA, offline PIN, bypass PIN allowed
Template 7	For quasi-credit cards, DDA, online PIN, bypass PIN allowed
Template 8	For Financial Social Security cards which are only available in Mainland China, DDA, online PIN, bypass PIN now allowed
Template 9	For signature debit cards which are only issued outside of Mainland China, DDA, Online PIN+signature, bypass PIN allowed
Template 10	For debit cards and certain pre-paid cards issued in the U.S., DDA, online PIN+signature, bypass PIN allowed
Template 11	For pure electronic cash (E-cash) cards
Template 12	For common E-cash cards, version 1
Template 13	For common E- cash cards, version 2
Template 14	For dual-currency E-cash cards
Template 15	For E-cash cards of extended applications
Template 16	For E-cash cards that support enhanced functionality
Template 17	For online only cards with no E-cash functions

Each template described here has a row of data tag, data object and tag value, which provide the stipulated functions. Card or mobile payment product (such as Apple Pay, Samsung Pay) can choose the template based on the product credit/debit attribute and risk preference. The UICS Debit/Credit application template 1, template 2, and template 9 can be used for prepaid template simultaneously. However, use of certain data objects are restricted by card type (card-specific) or special requirements of the Issuer (issuer-specific). These data objects are not listed in the templates.

Some examples of relevant special data objects are as follows:

AFL

Application File Locator (AFL) describes the file location and range of record where the data read by terminal during transaction processing are stored.

PSE vs. ADF –

If both the card and the terminal support optional Directory Selection Method feature, Payment System Environment (PSE) should be accessed. This file lists every payment application supported by the card.

Application Definition File (ADF) contains the data objects used during the transaction processing, which is a mandatory feature possessed by card and terminal.

Examples of issuer-specific data object are as follows:

File tag for the card in production cycle (tag 9F7F) – provide audit ability for card throughout the card life cycle.

Application Preferred Name (tag 9F12) – Preferred mnemonic name related to AID. If the terminal supports character sets specified by the issuer code table index, the name will appear.

Issuer Code Table Index (tag 9F11) – Code table displays the application preferred name.

The following data objects are described in the templates. Since they depend on issuer selection and requirements, values may vary.

CVM List (tag 8E) – Lists all cardholder verification methods supported by card application in accordance with the priority sequence.

Issuer Action Code (tags 9F0D, 9F0E, 9F0F) – 9F0D (default): Specifies transaction rejection conditions when the transaction requests to go online but the terminal could not finish the online submission; 9F0E (decline): specifies conditions for declining the transaction without online processing. 9F0F (online): specifies conditions for online processing.

Application Usage Control (tag 9F07) – indicates some issuer-specific restrictions on card applications, including geographic usage and service type, etc.

Frequency checking parameters – actual value of these variables should be set by the bank. These data include:

Consecutive Offline Transaction Limit (tag 9F53)

Cumulative Total Transaction Amount Limit (tag 9F54)

Lower Consecutive Offline Limit (tag 9F58)

Upper Consecutive Offline Limit (tag 9F59)

Cumulative Total Transaction Amount Upper Limit (tag 9F5C)

Consecutive Offline Transaction Limit (tag 9F72)

Cumulative Offline Transaction Amount Limit (tag 9F75)

Card Additional Processes (tag 9F68) – Labels the IC cards from the issuer that supports contactless payment for choice and support for transaction processes.

Card Transaction Qualifiers (tag 9F6C) –Labels the choice of transaction process in the event of offline data verification failure and the supported methods of cardholder verification for IC cards from the issuer that supports contactless payment.

4.1 Template 1 — Debit Card — DDA, Online PIN+Signature, Bypass PIN Allowed

4.1.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C 00 Refer to Section 4.1.2 for details	Describes the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.1.3 for details	If the card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Version Number	b	'9F08'	2	Initialized. 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its own version No. with that of the card.				
Card Internal Data (The table below summarizes the usage of these internal indicators.)	var.	-		Refer to Section 4.1.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence the transaction results.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates that the last online transaction hasn't been finished.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Online request by card indicator	b	-	1 bit	The initial setting is 0	During the transaction processing, set when the card decides transaction online.				
Offline decline requested by Card indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates the issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Labels whether SDA fails, when the last online transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction going online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of the last submission of online transaction				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records. <i>UICS</i> has provided the recommended values: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.1.5 for details	List data object tag and length in log record				
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommend value 00	A maximum number of consecutive offline transactions not using specified application currency. Requests online transaction when this number is exceeded				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommend value 00	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests online transaction when the counter exceeds the cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00 00 00 00 00 00	Records the cumulative offline transaction amount since the completion of last online transaction using currency specified by the application				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommend value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. The transaction requests to go online when this number is exceeded				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommend value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If exceeded and the transaction cannot go online, the transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommend value 00	Maximum consecutive offline transaction limit allowed by the card before applying for online authorization.				
Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommend value 00	Upper consecutive offline transaction limit allowed by the card before required by the issuer; when this value is exceeded, the transaction will be declined if the request to go online fails.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), the data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to Section 4.1.7 for details	Lists the data that the card requests the terminal to transmit of in second Generate AC. Its content is terminal data objects (tag and length), the data include: the issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time				
Cryptogram information data	b	'9F27'	1	The initial setting is 0	Indicates the type of cryptogram returned by the card				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer Action Code (IAC) - decline	b	'9F0E'	5	00 10 80 00 00 Refer to Section 4.1.8 for details	Specifies the condition that cause the transaction to be declined directly without going online.				
Issuer Action Code (IAC) - online	b	'9F0F'	5	D8 68 1C F8 00 Refer to 4.1.8 for details	Specifies the conditions for transaction online submission.				
Issuer Action Code (IAC) - default	b	'9F0D'	5	D0 60 1C A8 00 Refer to 4.1.8 for details	Specify conditions for transaction rejection when transaction request to go online but the terminal cannot finish online submission.				
Issuer Application Data	b	'9F10'	Var.	07 _ _ 01 03 00 00 00 01 Refer to Section 4.1.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer Country Code	n	'5F28'	2	Issuers fill in according to actual situation	Indicates the country where the card issuer is located Please refer to Appendix A.3				
Preferred Language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to the issuer preferred language.				
Application Currency Code	n	'9F51'	2	Issuers fill in according to actual situation	Domestic currency of the Issuer. Please refer to Appendix A.3				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Identifier (AID)	b	'4F'	5-16	Initialized Refer to Section 4.1.10 for details	UnionPay debit card AID: A0 00 00 03 33 01 01 01				
Application Label	ans	'50'	1-16	Filled in by Issuers according to the actual situation. Refer to Section 4.1.10 for details	Application name when the terminal displays an optional application list to the consumer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Usage Control	b	'9F07'	2	FF 00 Refer to Section 4.1.11 for details	Labels certain card-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				
Application Primary Account Sequence Number	n	'5F34'	1	Issuers fill in according to actual situation	Used to specify different applications using the same account in the card.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder Name	ans	'5F20'	2-26	Provided in magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element. Should not write the real full name of cardholder into the application				
Cardholder Name Extension	ans	'9F0B'	27-45	Issuers fill in according to actual situation	If cardholder name exceeds 26 bytes, the complete part is placed in this data element. Should not write the real full name of cardholder into the application				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID No. Should not write the real full ID No. of cardholder into the application				
Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicates cardholder ID type				
Application expiry date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiry date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.				
Application primary account No. (PAN)	cn	'5A'	10 at maximum	Provided by magnetic stripe data files	Equivalent to application primary account No. in magnetic stripe.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service codes defined in magnetic stripe.				
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				
Track 2 equivalent data	var.	'57'	19 at maximum	Provided by magnetic stripe data files	Equivalent to track 2 data (but the format is not same) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder Verification Method (CVM) list	b	'8E'	14	0000 0000 0000 0000 4203 1E03 1F00 Refer to Section 4.1.12 for details	Lists all CVMs supported by card application in order of priority. <i>Note: An application can have multiple CVM lists, for example: one is for domestic transactions and the other is for cross-border transactions.</i>				
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used together with RID to identify CA public key in the SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N _{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key index	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, and verified by the terminal in the SDA process.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Default Action (ADA)	b	'9F52'	2	C000 Refer to Section 4.1.13 for details	If issuer authentication is supported, They are UICS specialized data defining issuer specified actions carried out by the card under special conditions. All defaults set as 0 if this data does not exist in the card.				
Unique DEA Key (UDK) A	b	—	8	Issuers fill in according to actual situation	The Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (UDK) B	b	—	8	Issuers fill in according to actual situation	The Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (ENC Key)A	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key)B	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key)A	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (MAC Key)B	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				

4.1.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

4.1.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7-5	000	RFU
1	4-1	0001	Top priority

4.1.4 Card Internal Data

Card Internal Data	Reserved	Initial Value	Tag
Online authorization indicator	1 bit	0	-

Card Internal Data	Reserved	Initial Value	Tag
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit		-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 = mandatory)	1 byte	00 or 80, Recommended 00	'9F56'
Issuer script command counter	4 bits		-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation =0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation >= lower consecutive offline transaction limit, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'

Card Internal Data	Reserved	Initial Value	Tag
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation=0	'9F54'
Cumulative Total Transaction Amount Upper Limit	6 bytes	>=Cumulative total transaction amount limit (domestic), =0	'9F5C'

Note: To support SDA, issuer authentication and authorization control processing, indicator in the shadow shall be set during the personalization process.

4.1.5 Log Format

‘9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

4.1.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	20

4.1.7 Card Risk Management Data Object List 2

'8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization Response Code	8A	2
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3

4.1.8 Issuer Action Code (Decline, Online and Default)

'00 10 80 00 00' hexadecimal (IAC - decline)

‘D8 68 1C F8 00’ hexadecimal (IAC - online)

‘D0 60 1C A8 00’ hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline - Offline decline	IAC online - Request online	IAC default - Offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA fails		0	1	1
IC card data missing		0	0	0
Card appears in terminal exception file		0	1	1
DDA fails		0	1	0
CDA failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification fails	Byte 3	1	0	0

Unknown CVM		0	0	0
Number of PIN try exceeds the limit		0	0	0
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad present, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds the lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication fails		0	0	0

Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GAC command		0	0	0
RFU		0000	0000	0000

4.1.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm indicator
9	8-1		If issuer defined data present, self-defined data length (15 max)

4.1.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier(RID)	Proprietary Identifier Extension(PIX)	Card Type
A0 00 00 03 33	01 01 01	UnionPay Debit

4.1.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed .If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.1.12 Cardholder Verification Method List

‘0000 0000 0000 0000 4203 1E03 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0100 0010 0000 0011	Online PIN	1	If supported by the terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature	2	If supported by the terminal	Cardholder verification fails
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.1.13 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform transaction online next time.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	In case of a new card, don’t process online transaction.

Byte	Bit	Value	Definition
1	1	0	In case of new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	0	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is blocked in previous transaction, do not decline transaction or block application
2	2	0	Tag '9F79' / 'DF79' NOT reset during GAC processing
2	1	0	RFU

Note: Indicator in the shadow represents that it supports card and issuer authentication and authorization control processing.

4.2 Template 2 — Debit Card — DDA, Online PIN, Bypass PIN NOT Allowed

4.2.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 4.2.2 for details	Indicates the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.2.3 for details	If the card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its version No. with that of the card.				
Card Internal Data (next table summarizes usages of these internal indicators)	var.	-		Refer to Section 4.2.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction results.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates that the last online transaction is not finished.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Online transaction request indicator	b	-	1 bit	The initial setting is 0	During the transaction processing, set when card decides transaction online.				
Offline decline request indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides to decline transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Labels whether SDA fails when the last online transaction is rejected.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether DDA fails when the last online transaction is declined.				
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of last online transaction				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS</i> recommends value: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.2.5 for details	List data object tag and length in log record				
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 0	A maximum number of consecutive offline transactions not using specified application currency; the transaction requests to go online when this number is exceeded				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 0	A maximum number of consecutive offline transactions not in Issuer country; the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 0	International offline transaction counters. The card requests that transaction for online processing when counter exceeds cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 0	Records offline transaction cumulative amount since the completion of last online transaction using currency specified by the application				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. Requests to go online for transaction exceeding this limit				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If it is exceeded and the transaction cannot go online, the transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 0	Maximum consecutive offline transaction limit allowed by the card before applying for online authorization.				
Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 0	Upper consecutive offline limit allowed by the card before requesting online processing and specified by Issuer; when this value is exceeded, the transaction will be declined if the request to go online failed.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.2.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to 4.2.7 for details	List the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification results, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time.				
Cryptogram information data	b	'9F27'	1	The initial setting is 0	Indicates the type of cryptogram returned by the card				
Issuer Action Code (IAC) - decline	b	'9F0E'	5	00 10 98 00 00 Refer to 4.2.8 for details	Specifies the conditions for declining the transaction directly without going online..				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer Action Code (IAC) - online	b	'9F0F'	5	D8 68 04 F8 00 Refer to 4.2.8 for details	Specifies the conditions for transaction online submission.				
Issuer Action Code (IAC) - default	b	'9F0D'	5	D8 60 04 A8 00 Refer to 4.2.8 for details	Specify the conditions for transaction declining when the transaction requests to go online but terminal cannot complete online submission.				
Issuer application data	b	'9F10'	Var.	07_ _01 03 00 00 00 01Refer to 4.2.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Specifies the country of card issuer. Please refer to Appendix A.3				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.				
Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer domestic currency. Please refer to Appendix A.3				
Application Identifier (AID)	b	'4F'	5-16	Initialized Refer to Section 4.2.10 for details	UnionPay debit card AID: A0 00 00 03 33 01 01 01				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Label	ans	'50'	1-16	Filled in by Issuers according to the actual situation. Refer to Section 4.2.10 for details	Application name when the terminal displays an optional application list to the consumer.				
Application Usage Control	b	'9F07'	2	FF 00 Refer to 4.2.11 for details	Indicates certain issuer-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				
Application Primary Account Sequence Number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder Name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder Name Extension	ans	'9F0B'	27 - 45	Issuers fill in according to actual situation	If cardholder name has more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder ID Number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number				
Cardholder ID Type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicates cardholder ID type				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application expiry date	n	'5F24'	3	Provided by magnetic stripe data files	Card Application expiry date.				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.				
Application primary account No. (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account No. on magnetic stripe.				
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service codes defined in magnetic stripe.				
Track 1 Self-Defined Data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Track 2 Equivalent Data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (the format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				
Cardholder Verification Method (CVM) List	b	'8E'	12	0000 0000 0000 0000 0203 1F00 Refer to 4.2.12 for details	Lists all CVMs supported by card application in order of priority. Note: An application can have multiple CVM list, for example: one for domestic transactions and the other one for international transactions.				
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used together with RID to identify CA public key in the SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N _{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key exponent	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Used to verify the signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in the SDA process.				
IC card public key certificate	b	'9F46'	N_1	Issuers fill in according to actual situation	IC card public key authenticated by the Issuer.				
IC card public key exponent	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify the signature.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
IC card public key remainder	b	'9F48'	$N_{IC} - N_I + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.				
IC card private key	b	-	N_{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Applied to offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.				
Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 4.2.13 for details	Required data list sent to the card from the terminal in internal authentication command; including data object tag and length.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application default action (ADA)	b	'9F52'	2	C000 Refer to Section 4.2.14 for details	If issuer authentication is supported, they are UICS specialized data, to define issuer specified actions carried out by the card in special conditions. Defaults to all 0 if this data does not exist in the card.				
Unique DEA Key (UDK) A	b	—	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (UDK) B	b	—	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (ENC Key) A	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) B	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key) A	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key) B	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				

4.2.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

4.2.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation
1	7-5	000	RFU
1	4-1	0001	Top priority

4.2.4 Card Internal Data

Card Internal Data	RFU	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-

Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8: 0 = optional 1 = mandatory)	1 byte	00 or 80 Recommend 00	'9F5 6'
Issuer script command counter	4 bits		-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation =0	'9F5 8'
Upper consecutive offline transaction limit	1 byte	>=lower consecutive offline transaction limit, =0	'9F5 9'
Last online application transaction counter (ATC) register	2 bytes	0	'9F1 3'
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F5 3'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F7 2'
Cumulative offline transaction counter (international)	1 byte	0	-

PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation=0	'9F54'
Cumulative Total Transaction Amount Upper Limit	6 bytes	>=Cumulative offline transaction amount limit (domestic), =0	'9F5C'

Note: To support DDA, SDA, the issuer authentication and authorization control processing, indicator in shadow should be set during personalization.

4.2.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2

Data Object Name	Tag	Length
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

4.2.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	20

4.2.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization response code	8A	2
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3

4.2.8 Issuer Action Code (Decline, Online and Default)

'00 10 98 00 00' hexadecimal (IAC - decline)

'D8 68 04 F8 00' hexadecimal (IAC - online)

'D8 60 04 A8 00' hexadecimal (IAC - default)

Condition	Byte (Bit 8~Bit 1)	Result		
		IAC decline- Offline decline	IAC Online- Request Online	IAC default - Offline decline, if cannot go online
ODA not performed	Byte 1	0	1	1
SDA fails		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
DDA fails		0	1	1
CDA fails		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0

RFU		000	000	000
Cardholder verification fails	Byte 3	1	0	0
Unknown CVM		0	0	0
Number of PIN try exceeds the limit		0	0	0
PIN entry required, but PIN pad not present or not working		1	0	0
PIN entry required, PIN pad exists, but no PIN is entered		1	0	0
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds the lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000

Use default TDOL	Byte 5	0	0	0
Issuer authentication fails		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.2.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1		Length (max 15) of self-defined data by issuer, if any.

4.2.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier(RID)	proprietary identifier	Card Type

Application Identifier (AID)		Application Label
	extension(PIX)	
A0 00 00 03 33	01 01 01	UnionPay Debit

4.2.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed .If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.2.12 Cardholder Verification Method List

‘0000 0000 0000 0000 0203 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0000 0010 0000 0011	Online PIN	1	If supported by the terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	2	Always	Will not fail

4.2.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable number	9F37	4

4.2.14 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform next transaction online.

Byte	Bit	Value	Definition
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	In case of a new card, don't process online transaction.
1	1	0	In case of new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	0	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is blocked in previous transaction, do not decline transaction or block application
2	2	0	Tag '9F79' / 'DF79' NOT reset during GAC processing
2	1	0	RFU

Note: Shaded indicator means it supports card and issuer authentication and authorization control processing.

4.3 Template 3 — Credit Card — DDA, Online PIN, Bypass PIN Allowed

4.3.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 4.3.2 for details	Indicate the functions supported by card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.3.3 for details	If card has multiple applications, indicate application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. Terminal will compare its version No. with that of card.				
Card Internal Data (next table summarizes usage of these internal indicators)	var.	-		Refer to Section 4.3.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction results.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicate the last online transaction is incomplete.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Online transaction request indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when card decides transaction online.				
Offline decline request indicator	b	-	1 bit	The initial setting is 0	During the process of transaction processing, set when card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicate issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicate whether SDA failed when the last online transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicate whether DDA failed when the last online transaction is declined.				
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of the last online transaction				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS Specification</i> has provided the recommendation values: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.3.5 for details	List data object tag and length in log record				
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 0	A maximum number of consecutive offline transactions not using specified application currency; the transaction requests to go online when this number is exceeded				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Consecutive offline transaction limit (international - currency)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 0	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 0	International offline transaction counters. The card requests the transaction for online processing when counter exceeds cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 0	Records offline transaction cumulative amount since the completion of last online transaction using currency specified by the application				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. Requests to go online for transaction exceeding this limit				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If it is exceeded and the transaction cannot go online, the transaction declined.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
lower consecutive transaction offline limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 0	Maximum consecutive offline transaction limit allowed by card before applying for online authorization.				
upper consecutive transaction offline limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 0	Upper consecutive offline limit allowed by the card before requesting online processing and specified by Issuer; when this value is exceeded, the transaction will be declined if the request to go online failed.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.3.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to 4.3.7 for details	List the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time				
Cryptogram information data	b	'9F27'	1	The initial setting is 0	Indicate the type of cryptogram returned by the card				
Issuer Action Code (IAC) - decline	b	'9F0E'	5	00 10 00 00 00 Refer to 4.3.8 for details	Specify the conditions for declining the transaction directly without going online.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Issuer Action Code (IAC) - online	b	'9F0F'	5	D8 68 9C F8 00 Refer to 4.3.8 for details	Specify the conditions for online transaction online submission.				
Issuer Action Code (IAC) - default	b	'9F0D'	5	D8 60 9C A8 00 Refer to 4.3.8 for details	Specify conditions for transaction declining when the transaction requests to go online but terminal cannot complete online submission.				
Issuer application data	b	'9F10'	Var.	07 _ _ 01 03 00 00 00 01 Refer to 4.3.9 for details	Specific application data that need to be sent to the Issuer in an online transaction. (One Derived Key Index (DKI))				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicate the country of card issuer. Please refer to Appendix A.3				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.				
Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer's domestic currency. Please refer to Appendix A.3				
Application Identifier(AID)	b	'4F'	5-16	Initialized Refer to Section 4.3.10 for details	Registered Application Provider Identifier (RID) and proprietary identifier extension: A0 00 00 03 33 01 01 02 A0 00 00 03 33 determines UICS registered application provider (same for all cards), 01 01 02 indicates UICS credit application.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Application Label	ans	'50'	1-16	Fill in by Issuers according to actual situation. Refer to Section 4.3.10 for details	Application name when the terminal displays an optional application list to the consumer.				
Application Usage Control	b	'9F07'	2	FF 00 Refer to 4.3.11 for details	Indicates certain issuer-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.				
Cardholder name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder name extension	ans	'9F0B'	27-45	Issuers fill in according to actual situation	If cardholder name is more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicate Cardholder ID type				
Application expiry date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiry date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.				
Application primary account No. (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account No. on magnetic stripe.				
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service code defined in magnetic stripe.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				
Track 2 equivalent data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (the format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				
Cardholder verification method (CVM) list	b	'8E'	14	0000 0000 0000 0000 4203 1E03 1F00 Refer to 4.3.12 for details	Lists all CVMs supported by card application in order of priority. Note: an application can have multiple CVM list, for example: one for domestic transaction and the other one for cross-border transaction.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used with RID to identify CA public key in SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N_{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.				
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key exponent	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in SDA process.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
IC card public key certificate	b	'9F46'	N_I	Issuers fill in according to actual situation	IC card public key authenticated by the Issuer.				
IC card public key exponent	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify signature.				
IC card public key remainder	b	'9F48'	$N_{IC} - N_I + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.				
IC card private key	b	-	N_{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Applied to offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 4.3.13 for details	Data list that need to be sent to the card from the terminal in internal authentication command, including data object tag and length.				
Application default action (ADA)	b	'9F52'	2	C000 Refer to Section 4.3.14 for details	If issuer authentication is supported. UICS specialized data. Defines issuer specified actions carried out by card in special conditions. Defaults to all 0 if this data does not exist in the card.				
Unique DEA Key (UDK) A	b	—	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Unique DEA Key (UDK) B	b	—	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) A	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) B	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key) A	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (Bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Special Data	Data Stored in File Record
Unique DEA Key (MAC Key) B	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				

4.3.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

4.3.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7-5	000	RFU
1	4-1	0001	Top priority

4.3.4 Card Internal Data

Card Internal Data	RFU	Initial Value	Tag
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Card Internal Data	RFU	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit 0 = optional 1 = mandatory)	1 byte	00 or 80 Recommend 00	'9F56'
Issuer script command counter	4 bits	00	-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation = 0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation ≥ LCOL, = 0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation = 0	'9F53'

Card Internal Data	RFU	Initial Value	Tag
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation = 0	'9F54'
Cumulative Total Transaction Amount Upper Limit	6 bytes	Issuers fill in according to actual situation > = CTAL, =0	'9F5C'

Note: To support DDA, SDA, the issuer authentication and authorization control processing, indicator in shadow should be set during personalization.

4.3.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

4.3.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3

Data Object Name	Tag	Length
Merchant name and location	9F4E	20

4.3.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization response code	8A	2
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3

4.3.8 Issuer Action Code (Decline, Online and Default)

‘00 10 00 00 00’ hexadecimal (IAC - decline)

‘D8 68 9C F8 00’ hexadecimal (IAC - online)

‘D8 60 9C A8 00’ hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline – offline decline	IAC online – Request online	IAC default – Offline decline if cannot go online
Offline data authentication not performed	Byte 1	0	1	1
Offline static data authentication (SDA) failed		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
Offline dynamic data authentication (DDA) failed		0	1	1
Combined DDA/Application Cryptogram generation (CDA) failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0

New card		0	1	0
RFU		000	000	000
Cardholder verification failed	Byte 3	0	1	1
Unknown CVM		0	0	0
Number of PIN try exceeded the limit		0	0	0
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad exists, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant request to go online processing		0	1	1
RFU		000	000	000

Use default TDOL	Byte 5	0	0	0
Issuer authentication failed		0	0	0
Script processing failed before the final GENERATE AC command failed		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.3.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1		If issuer self-defined data exist, self-defined data length (15 maximum.)

4.3.10 Application Identifier and Application Label

Application Identifier (AID)	Application Label
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Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	proprietary identifier extension (PIX)	Card Type
A0 00 00 03 33	01 01 02	UnionPay Credit

4.3.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back allowed
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed. If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.3.12 Cardholder Verification Method List

‘0000 0000 0000 0000 4203 1E03 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Conditions	Action if CVM Fails
0100 0010 0000 0011	Online PIN	1	If supported by terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature (on paper)	2	If supported by terminal	Fail Cardholder verification

CVM code – first eight bytes are 0	CVM	Processing Order	Conditions	Action if CVM Fails
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.3.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable number	9F37	4

4.3.14 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform next transaction online.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	In case of a new card, don't process online transaction.
1	1	0	In case of new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.

Byte	Bit	Value	Definition
2	7	0	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is blocked in previous transaction, do not decline transaction or block application
2	2	0	Tag 9F79/DF79 NOT reset during GAC processing
2	1	0	RFU

Note: Indicator bit in shadow represented supports card and issuer authentication and authorization control processing.

4.4 Template 4 — Credit Card — DDA, Signature, Online Control

4.4.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 4.4.2 for details	Indicate the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.4.3 for details	If the card has multiple applications, indicate application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. Terminal will compare its version No. with that of the card.				
Card Internal Data (next table summarizes usage of these internal indicators)	var.	-		Refer to Section 4.4.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction result.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicate that last online transaction is not finished.				
Online request by card indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides transaction online.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Offline decline requested by card indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicate issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether SDA failed when the last online transaction was declined.				
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicate whether DDA failed when the last online transaction is rejected.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of last online transaction				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS</i> has provided the recommended values: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.4.5 for details	List data object tag and length in log record				
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 00	A maximum number of consecutive offline transactions not using specified application currency; the transaction requests to go online when this number is exceeded				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 00	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests that transaction gets online when the counter exceeds cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00	Records cumulative offline transaction amount since the completion of last online transaction using currency specified by the application				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. The transaction requests to go online when this number is exceeded				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If exceeded and the transaction cannot go online, the transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 00	Maximum consecutive offline transaction limit allowed by card before applying for online authorization.				
upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 00	Upper consecutive offline transaction limit allowed by the card before requesting online processing and specified by Issuer; when this value is exceed, the transaction will be declined if the request to go online failed.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List 1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.4.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List 2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to 4.4.7 for details	List the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time.				
Cryptogram information data	b	'9F27'	1	The initial setting is 00	Indicate the type of cryptogram returned by the card				
Issuer action code (IAC) - decline	b	'9F0E'	5	00 10 00 00 00 Refer to 4.4.8 for details	Specify the conditions for declining the transaction directly without going online.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer action code (IAC) - online	b	'9F0F'	5	D8 68 04 F8 00 Refer to 4.4.8 for details	Specify the conditions for transaction online submission.				
Issuer action code (IAC) - default	b	'9F0D'	5	D8 60 04 A8 00 Refer to 4.4.8 for details	Specify conditions for transaction declining when the transaction request to go online but terminal cannot complete online submission.				
Issuer application data	b	'9F10'	Var.	07__ 01 03 00 00 00 01 Refer to 4.4.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicate the country of card issuer. Please refer to Appendix A.3				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.				
Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer domestic currency. Please refer to Appendix A.3				
Application Identifier(AID)	b	'4F'	5-16	Initialized Refer to Section 4.4.10 for details	Registered Application Provider Identifier (RID) and proprietary identifier extension: A0 00 00 03 33 01 01 02 A0 00 00 03 33 determine UICS registered application provider (same for all cards), 01 01 02 indicate UICS credit application.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application label	ans	'50'	1-16	Issuers fill in according to actual situation Refer to Section 4.4.10 for details	Application name when the terminal displays an optional application list to the consumer.				
Application Usage Control	b	'9F07'	2	FF 00 Refer to 4.4.11 for details	Indicate certain issuer-specific restrictions on card application specified, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder name extension	ans	'9F0B'	27 - 45	Issuers fill in according to actual situation	If cardholder name is more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number				
Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicate Cardholder ID type				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application expiry date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiry date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.				
Application primary account No. (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account No. on magnetic stripe.				
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service code defined in magnetic stripe.				
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Track 2 equivalent data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (the format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				
Cardholder verification method (CVM) list	b	'8E'	14	0000 0000 0000 0000 1E03 0203 1F00 Refer to 4.4.12 for details	List all CVMs supported by card application in order of priority. Note: an application can have multiple CVM list, for example: one for domestic transactions and the other one for international transactions.				
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used with RID to identify CA public key in SDA or DDA process.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer public key (IPK) certificate	b	'90'	N_{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.				
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key exponent	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in SDA process.				
IC card public key certificate	b	'9F46'	N_1	Issuers fill in according to actual situation	IC card public key authenticated by the Issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
IC card public key exponent	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify signature.				
IC card public key remainder	b	'9F48'	$N_{IC} - N_I + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.				
IC card private key	b	-	N_{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Applied to offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 4.4.13 details	Data list that need to be sent to the card from the terminal in internal authentication command; including data object tag and length.				
Application default action (ADA)	b	'9F52'	2	C000 Refer to Section 4.4.14 for details	If issuer authentication is supported, they are UICS specialized data to define issuer specified actions carried out by the card in special conditions. Defaults to all 0 if this data does not exist in the card.				
Unique DEA Key (UDK) A	b	—	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (UDK) B	b	—	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) A	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) B	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key) A	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (MAC Key) B	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				

4.4.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	00 00 00 00	RFU

4.4.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7-5	00 0	RFU
1	4-1	00 01	Top priority

4.4.4 Card Internal Data

Card Internal Data	RFU	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 = mandatory)	1 byte	00 or 80, Recommend 00	'9F56'
Issuer script command counter	4 bits	00	-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation = 0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation >= LCOL, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'

Card Internal Data	RFU	Initial Value	Tag
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation = 0	'9F54'
Upper cumulative offline transaction limit	6 bytes	Issuers fill in according to actual situation >= CTAL, =0	'9F5C'

Note: To support DDA, SDA, the issuer authentication and authorization control processing, indicator in shadow should be set during personalization.

4.4.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Authorized amount	9F02	6

Data Object Name	Tag	Length
Other amounts	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

4.4.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	20

4.4.7 Card Risk Management Data Object List 2

'8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization response code	8A	2
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3

4.4.8 Issuer Action Code (Decline, Online and Default)

'00 10 00 00 00' hexadecimal (IAC - decline)

'D8 68 04 F8 00' hexadecimal (IAC - online)

'D8 60 04 A8 00' hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline – offline decline	IAC online – Request online	IAC default – offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA failed		0	1	1
IC card data are missing		0	0	0
Card appeared in terminal exception file		0	1	1
DDA failed		0	1	1
CDA failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification failed	Byte 3	0	0	0

Unknown CVM		0	0	0
Number of PIN try exceeded the limit		0	0	0
PIN entry required, but PIN pad not present or not working		0	0	0
PIN entry required, PIN pad exists, but no PIN is entered		0	0	0
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication failed		0	0	0

Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.4.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1		Length (15 maximum) of self-define data by issuer, if any.

4.4.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier(RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 02	UnionPay Credit

4.4.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed. If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.4.12 Cardholder Verification Method List

‘0000 0000 0000 0000 1E03 0203 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0001 1110 0000 0011	Signature	1	If supported by terminal	Fail Cardholder Verification
0000 0010 0000 0011	Online PIN	2	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.4.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable number	9F37	4

4.4.14 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, process next transaction online.
	7	1	If issuer authentication executed but failed, decline transaction.
	6	0	If issuer authentication is necessary but does not receive ARPC, do not decline the transaction.
	5	0	If transaction is declined, do not generate notification.
	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
	3	0	If transaction is declined because issuer authentication failed or is not executed, do not generate notification.
	2	0	In case of a new card, don't process online transaction.
	1	0	In case of a new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
	7	0	If PIN is blocked in last transaction, do not decline transaction.
	6	0	If PIN is blocked in last transaction, don't process online transaction.
	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
	4	0	If issuer script command failed in last transaction, don't process online transaction.
	3	0	If PIN is blocked in previous transaction, do not decline transaction or block application
	2	0	Tag '9F79' / 'DF79' NOT reset in GAC processing

2	1	0	RFU
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Note: Indicator in shadow supports card and issuer authentication and authorization control processing.

4.5 Template 5 — Credit Card — DDA, Online PIN, ByPass PIN NOT Allowed

4.5.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 4.5.2 for details	Indicates the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.5.3 for details	If card has multiple applications, indicate application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				
Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its version No. with that of the card.				

Card Internal Data (next table summarizes usage of these internal indicators)	var.	-		Refer to Section 4.5.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction result.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates that last online transaction is not finished.				
Online transaction request indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides transaction online.				
Offline decline requested by card indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides to decline transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates issuer authentication error occurred in last transaction.				

Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether SDA failed when the last online transaction was declined.				
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether DDA failed when the last online transaction was declined.				
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				

Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of last online transaction				
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS Specification</i> has provided the recommendation values: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.5.5 for details	List data object tag and length in log record				
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 00	A maximum number of consecutive offline transactions not using specified application currency; the transaction requests to go online when this number is exceeded				

Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 00	A maximum number of consecutive offline transactions not in Issuer country; the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests that transaction for online processing when counter exceeds cumulative offline transaction limit.				
PIN try limit	b	-	1	Issuers fill in according to actual situation	Number of incorrect PIN try cardholder can try.				
PIN try number counter	b	'9F17'	1	The initial setting is 0, set as PIN try limit during personalization	Equal to PIN try limit.				

Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00 00 00 00 00 00	Records offline transaction cumulative amount since the completion of last online transaction using currency specified by the application				
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. The transaction requests to go online when this number is exceeded				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If it is exceeded and the transaction cannot go online, the transaction is declined.				

Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 00	Maximum consecutive offline transaction limit allowed by card before applying for online authorization.				
Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 00	Upper consecutive offline limit allowed by the card before requesting online processing, and specified by issuer; when this value is exceed, the transaction will be declined if the request to go online failed.				

Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.5.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				
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Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to 4.5.7 for details	List the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time.				
Cryptogram information data	b	'9F27'	1	The initial setting is 00	Indicate the type of cryptogram returned by the card				
Issuer action code (IAC) - decline	b	'9F0E'	5	00 10 98 00 00 Refer to 4.5.8 for details	Specify the conditions for declining the transaction directly without going online.				
Issuer action code (IAC) - online	b	'9F0F'	5	D8 68 04 F8 00 Refer to 4.5.8	Specify the conditions for transaction online submission.				

Issuer action code (IAC) - default	b	'9F0D'	5	D8 60 04 A8 00 Refer to 4.5.8 for details	Specify conditions for transaction declining when the transaction request to go online but terminal cannot complete online submission.				
Issuer application data	b	'9F10'	Var.	07_ _01 03 00 00 00 01 Refer to 4.5.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicate the country of card issuer. Please refer to Appendix A.3				
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.				
Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer domestic currency. Please refer to Appendix A.3				

Application Identifier(AID)	b	'4F'	5-16	Initialized Refer to Section 4.5.10 for details	Registered Application Provider Identifier (RID) and proprietary identifier extension: A0 00 00 03 33 01 01 02 A0 00 00 03 33 determines UICS registered application provider (same for all cards), 01 01 02 indicates UICS credit application.				
Application label	ans	'50'	1-16	Fill in by Issuers according to the actual situation Refer to Section 4.5.10 for details	Application name when the terminal displays an optional application list to the consumer.				

Application Usage Control	b	'9F07'	2	FF 00 Refer to 4.5.11 for details	Indicate certain issuer-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.				
Cardholder name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe.				
Cardholder name extension	ans	'9F0B'	27 - 45	Issuers fill in according to actual situation	If cardholder name is more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number				

Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicates Cardholder ID type				
Application expiry date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiry date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.				
Application primary account No. (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account No. on magnetic stripe.				
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service code defined in magnetic stripe.				
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				

Track 2 equivalent data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (the format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				
Cardholder verification method (CVM) list	b	'8E'	14	0000 0000 0000 0000 0203 1E03 1F00 Refer to 4.5.12 for details	Lists all CVMs supported by card application in order of priority. <i>Note: an application can have multiple CVM list, for example: one for domestic transactions and the other for cross-border transactions.</i>				
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used with RID to identify CA public key in SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N_{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.				
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				

Issuer public key exponent	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in SDA process.				
IC card public key certificate	b	'9F46'	N_1	Issuers fill in according to actual situation	IC card public key authenticated by issuer.				
IC card public key exponent	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify signature.				
IC card public key remainder	b	'9F48'	$N_{IC} - N_1 + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.				

IC card private key	b	-	N _{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Applied to offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.				
Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 4.5.13 for details	Data list that need to be sent to the card from the terminal in internal authentication command; including data object tag and length.				
Application default action (ADA)	b	'9F52'	2	C040 Refer to Section 4.5.14 for details	If issuer authentication is supported, they are UICS specialized data, to define issuer specified actions carried out by the card in special conditions. Defaults to all 0 if this data does not exist in the card.				
Offline PIN	cn	—	2-6	Provided by issuer	Write to the card by issuer during personalization				

Unique DEA Key (UDK) A	b	–	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (UDK) B	b	–	8	Issuers fill in according to actual situation	Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) A	b	–	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key) B	b	–	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key) A	b	–	8	Issuers fill in according to actual situation	Secure message key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Unique DEA Key (MAC Key) B	b	_	8	Issuers fill in according to actual situation	Secure message key applied to issuer script, Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				

4.5.2 Application Interchange Profile

‘7C00’ hexadecimal

B y te	B i t	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8 - 1	0000 0000	RFU

4.5.3 Application Priority Indicator

‘01’ hexadecimal

B yt e	B i t	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7 - 5	000	RFU
1	4 - 1	0001	Top priority

4.5.4 Card Internal Data

Card Internal Data	RF U	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 = compulsory)	1 byte	00 or 80 Recommend 00	'9F56'
Issuer script command counter	4 bits	00	-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation = 0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation >= LCO L, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'

Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation = 0	'9F54'
Upper cumulative offline transaction limit	6 bytes	Issuers fill in according to actual situation >= CTAL, =0	'9F5C'

Note: to support DDA, SDA, offline PIN, the issuer authentication and authorization control processing, indicator bit in shadow should be set during personalization.

4.5.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Authorized amount	9F02	6
Other amount	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

4.5.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable numeric	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	20

4.5.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization response code	8A	2
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable numeric	9F37	4
Transaction time	9F21	3

4.5.8 Issuer Action Code (Decline, Online and Default)

'00 10 98 00 00' hexadecimal (IAC - decline)

'D8 68 04 F8 00' hexadecimal (IAC - online)

'D8 60 04 A8 00' hexadecimal (IAC - default)

Condition	Byte (Bit8~Bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline reject if cannot go online
Did not perform offline data authentication	Byte 1	0	1	1
Offline static data authentication (SDA) fails		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
Offline dynamic data authentication (DDA) fails		0	1	1
Combined DDA/Application Cryptogram generation (CDA) failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0

RFU		000	000	000
Cardholder verification fails	Byte 3	1	0	0
Unknown CVM		0	0	0
Number of PIN try exceeded the limit		0	0	0
PIN entry required, but PIN pad not present or not working		1	0	0
PIN entry required, PIN pad exists, but no PIN is entered		1	0	0
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant request online processing		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication fails		0	0	0

Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.5.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1		Length (15 Max) of self-defined data by Issuer, if any.

4.5.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 02	UnionPay Credit

4.5.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back allowed
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed .If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.5.12 Cardholder Verification Method List

‘0000 0000 0000 0000 0203 1E03 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0000 0010 0000 0011	Online PIN	1	If supported by terminal	Fail Cardholder Verification
0001 1110 0000 0011	Signature (on paper)	2	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.5.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable numeric	9F37	4

4.5.14 Application Default Action

‘C040’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If Issuer authentication fails, perform next transaction online.
1	7	1	If Issuer authentication executed but failed, decline the transaction.
1	6	0	If Issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication failed or is not performed, do not generate notification.
1	2	0	If it is a new card, don't process online transaction.
1	1	0	If it is a new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	1	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If Issuer script command failed in this transaction, don't process online transaction.
2	3	0	If PIN is locked in previous transaction, do not decline transaction or block application
2	2	0	Tag '9F79' / 'DF79' NOT reset during GAC processing
2	1	0	RFU

Note: Shaded indicator means it supports PIN, card and issuer authentication and authorization control processing.

4.6 Template 6 — Credit Card — CDA, Offline PIN, ByPass PIN Allowed

4.6.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Application Interchange Profile (AIP)	b	'82'	2	7D00 Refer to Section 4.6.2 for details	Indicates the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.6.3 for details	If card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				
Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its version No. with that of the card.				

Card Internal Data (next table summarizes usage of these internal indicators)	var.	-		Refer to Section 4.6.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction result.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates last online transaction incomplete.				
Online request by card indicator	b	-	1 bit	The initial setting is 0	In transaction processing, set when card decides transaction online.				
Offline decline requested by card indicator	b	-	1 bit	The initial setting is 0	In transaction processing, set when card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates that issuer authentication error occurred in last transaction.				

Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether SDA failed when the last online transaction was rejected.				
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether DDA failed when the last online transaction was declined.				
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or compulsory ('80'). If it is compulsory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of last online transaction				

Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS Specification</i> has provided recommendation value: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.6.5 for details	List data object tag and length in log records				

Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 00	Maximum number of consecutive offline transactions not using specified application currency; the transaction requests to go online when this number is exceeded				
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 00	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests transaction for online processing when counter exceeds cumulative offline transaction limit.				
Offline PIN	cn	-	2-6	Provided by Issuer	Write to the card by issuer during personalization				
PIN try limit	b	-	1	Issuers fill in according to actual situation	Number of incorrect PIN try cardholder can try.				

PIN try number counter	b	'9F17'	1	Initialized as 00, set as PIN try limit during personalization	Equal to PIN try limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00 00 00 00 00 00	Records offline transaction cumulated amount since the completion of last online transaction using currency specified by the application				
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. Requests online for transaction exceeding this limit				

Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If it is exceeded and the transaction cannot go online, the transaction is declined.				
Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 00	Maximum consecutive offline transaction limit allowed by card before applying for online authorization.				
Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 00	Upper consecutive offline limit allowed by the card before requesting online processing and specified by the Issuer; when this value is exceed, the transaction will be declined if the request to go online failed.				

Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.6.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorize control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				
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Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to 4.6.7 for details	Lists the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time.				
Cryptogram information data	b	'9F27'	1	The initial setting is 0	Indicates the type of cryptogram returned by card				
Issuer action code (IAC) - decline	b	'9F0E'	5	00 10 80 00 00 Refer to Section 4.6.8 for details	Specifies the conditions for declining the transaction directly without going online.				

Issuer action code (IAC) - online	b	'9F0F'	5	D8 68 3C F8 00 Refer to Section 4.6.8 for details	Specifies the conditions for transaction online submission.				
Issuer action code (IAC) - default	b	'9F0D'	5	D8 60 3C A8 00 Refer to Section 4.6.8 for details	Specifies conditions for transaction declining when the transaction requests for online processing but the terminal cannot complete online submission.				
Issuer application data	b	'9F10'	Var.	Refer to 4.6.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicates the country of card issuer. Please refer to Appendix A.3				
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.				

Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer domestic currency. Please refer to Appendix A.3				
Application Identifier(AID)	b	'4F'	5-16	Initialized Refer to Section 4.6.10 for details	Registered Application Provider Identifier (RID) and proprietary identifier extension: A0 00 00 03 33 01 01 02 A0 00 00 03 33 determines UICS registered application provider (same for all cards), 01 01 02 indicate UICS credit application.				
Application label	ans	'50'	1-16	Issuers fill in according to actual situation Refer to Section 4.6.10 for details	Application name when the terminal displays an optional application list to the consumer.				

Application Usage Control	b	'9F07'	2	FF 00 Refer to 4.6.11 for details	Indicate certain issuer-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.				
Cardholder name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder name extension	ans	'9F0B'	27 - 45	Issuers fill in according to actual situation	If cardholder name is more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number				

Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicates Cardholder ID type				
Application expiration date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiration date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in card.				
Application primary account number (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account number on magnetic stripe.				
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service code defined in magnetic stripe.				
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				

Track 2 equivalent data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				
Cardholder verification method (CVM) list	b	'8E'	16	0000 0000 0000 0000 4103 1E03 0203 1F00 Refer to 4.6.12 for details	Lists all CVMs supported by card application in order of priority.. Note: an application can have multiple CVM list, for example: one for domestic transaction and the other for international transaction.				
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used with RID to identify CA public key in SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N _{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Used for offline data authentication.				

Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key exponent	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in SDA process.				
IC card public key certificate	b	'9F46'	N_1	Issuers fill in according to actual situation	IC card public key authenticated by the Issuer.				
IC card public key exponent	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify signature.				
IC card public key remainder	b	'9F48'	$N_{IC} - N_1 + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.				

IC card private key	b	-	N _{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Used for offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.				
Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 4.6.13 for details	Data list that need to be sent to the card from the terminal in internal authentication command; including data object tag and length.				
Application default action (ADA)	b	'9F52'	2	C040 Refer to Section 4.6.14 for details	If issuer authentication is supported, they are UICS specialized data, to define issuer specified actions executed by the card in special conditions. Defaults to all 0 if this data does not exist in card.				
Offline PIN	cn	-	2-6	Provided by issuer	Write to the card by issuer during personalization				

Unique derivation key (UDK) A	b	—	8	Issuers fill in according to actual situation	Unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (UDK) B	b	—	8	Issuers fill in according to actual situation	Unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (ENC Key) A	b	—	8	Issuers fill in according to actual situation	Enciphered key used for issuer script; unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (ENC Key) B	b	—	8	Issuers fill in according to actual situation	Enciphered key used for issuer script; unique derivation key of each card is generated by the unique master derivation key unique to each issuer.				
Unique derivation key (MAC Key) A	b	—	8	Issuers fill in according to actual situation	Secure message key used for issuer script, unique derivation key of each card is generated by the unique master derivation key of each issuer.				

Unique derivation key (MAC Key) B	b	_	8	Issuers fill in according to actual situation	Secure message key used for issuer script, unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				

4.6.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA supported
2	8-1	0000 0000	RFU

4.6.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7-5	000	RFU
1	4-1	000 1	Top priority

4.6.4 Card Internal Data

Card Internal Data	RFU	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8: 0 = optional 1 = mandatory)	1 byte	00 or 80 Recommended 00	'9F56'
Issuer script command counter	4 bits	00	-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation = 0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation > =LCOL, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'

Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation = 0	'9F54'
Upper cumulative offline transaction limit	6 bytes	Issuers fill in according to actual situation >= CTAL, =0	'9F5C'

Note: To support DDA, SDA, offline PIN, the issuer authentication and authorization control processing, shaded indicator should be set during personalization.

4.6.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction Date	9A	3
Transaction Time	9F21	3
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Transaction Currency Code	5F2A	2
Merchant Name and Location	9F4E	20
Transaction Type	9C	1
Application Transaction Counter (ATC)	9F36	2

4.6.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	20

4.6.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization response code	8A	2
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3

4.6.8 Issuer Action Code (Decline, Online and Default)

'00 10 80 00 00' hexadecimal (IAC - decline)

'D8 68 3C F8 00' hexadecimal (IAC - online)

'D8 60 3C A8 00' hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline– Offline decline	IAC online – Request online	IAC default – Offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA fails		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
DDA fails		0	1	1
CDA failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
CVM fails	Byte 3	1	0	0
Unknown CVM		0	0	0

Number of PIN try exceeded the limit		0	1	1
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad exists, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant request online processing		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication fails		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0

Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.6.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1		Length (15 Max) of self-defined data by Issuer, if any.

4.6.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 02	UnionPay Credit

4.6.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
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Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back allowed
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed .If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.6.12 Cardholder Verification Method list

‘0000 0000 0000 0000 4103 1E03 0203 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0100 0011 0000 0011	Offline PIN	1	If supported by terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature (on paper)	2	If supported by terminal	Fail Cardholder Verification
0000 0010 0000 0011	Online PIN	3	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	4	Always	Will not fail

4.6.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable Number	9F37	4

4.6.14 Application Default Action

‘C040’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform next transaction online.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and the transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication failed or is not performed, do not generate notification.
1	2	0	If it is a new card, don't process online transaction.
1	1	0	If it is a new card, do not decline transaction when the transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	1	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when the transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is blocked in the previous transaction, do not decline transaction or block application
2	2	0	Tag '9F79' / 'DF79' NOT reset during GAC processing
2	1	0	RFU

Note: Shaded indicator means it supports PIN, card and issuer authentication and authorization control processing.

4.7 Template 7 — Quasi Credit Card — DDA, Online PIN, Bypass PIN Allowed

4.7.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 4.7.2 for details	Indicates the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.7.3 for details	If card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				
Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its version No. with that of the card.				

Card Internal Data (next table summarizes usage of these internal indicators)	var.	-		Refer to Section 4.7.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction result.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates last online transaction incomplete.				
Online request by card indicator	b	-	1 bit	The initial setting is 0	In transaction processing, set when card decides transaction online.				
Offline decline requested by card indicator	b	-	1 bit	The initial setting is 0	In transaction processing, set when the card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates that issuer authentication error occurred in last transaction.				

Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether SDA failed when the last online transaction was rejected.				
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether DDA failed when the last online transaction was declined.				
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or compulsory ('80'). If it is compulsory but authorization response cryptogram is not returned, the Issuer can choose to reject this transaction irrespective the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of last online transaction				

Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS</i> has provided recommendation value: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.7.5 for details	Lists data object tag and length in log records				

Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 0	Maximum number of consecutive offline transactions not using specified application currency, the transaction requests to go online when this number is exceeded				
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 0	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 0	International offline transaction counters. The card requests transaction for online processing when counter exceeds cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 0	Records offline transaction cumulated amount since the completion of last online transaction using currency specified by the application				

Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. Request online for transaction exceeding this limit				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If it is exceeded and the transaction cannot go online, the transaction is declined.				

Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 00	Maximum consecutive offline transaction limit allowed by card before applying for online authorization.				
Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 00	Upper consecutive offline limit allowed by the card before requesting online processing, which is specified by the Issuer; when this value is exceeded, the transaction will be declined if the request to go online failed.				

Card Risk Management Data Object List 1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.7.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				
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Card Risk Management Data Object List 2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to Section 4.7.7 for details	List the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time.				
Cryptogram information data	b	'9F27'	1	The initial setting is 00	Indicate the type of cryptogram returned by card				
Issuer action code (IAC) - decline	b	'9F0E'	5	00 10 80 00 00 Refer to Section 4.7.8 for details	Specifies the conditions for declining the transaction directly without going online.				

Issuer action code (IAC) - online	b	'9F0F'	5	D8 68 1C F8 00 Refer to Section 4.7.8 for details	Specifies the conditions for transaction online submission.				
Issuer action code (IAC) - default	b	'9F0D'	5	D8 60 1C A8 00 Refer to Section 4.7.8 for details	Specifies Conditions for transaction declining when the transaction request online but terminal cannot complete online escalation.				
Issuer application data	b	'9F10'	Var.	Refer to Section 4.7.9 for details	Specific application data that need to be sent to the Issuer in an online transaction. (A derived key index (DKI))				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicate the country of card issuer. Please refer to Appendix A.3				
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.				

Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer domestic currency. Please refer to Appendix A.3				
Application Identifier(AID)	b	'4F'	5-16	Initialized Refer to Section 4.7.10 for details	Registered Application Provider Identifier (RID) and proprietary identifier extension: A0 00 00 03 33 01 01 03 A0 00 00 03 33 determines UICS registered application provider (same for all cards), 01 01 03 indicates UICS quasi credit application.				
Application label	ans	'50'	1-16	Issuers fill in according to actual situation Refer to Section 4.7.10 for details	Application name when the terminal displays an optional application list to the consumer.				

Application Usage Control	b	'9F07'	2	FF 00 Refer to Section 4.7.11 for details	Indicates certain issuer-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code). '2' = hex FF00 is template default value, card set as '6' = hex AB00 when card only supports domestic functions.				
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.				
Cardholder name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.				
Cardholder name extension	ans	'9F0B'	27 - 45	Issuers fill in according to actual situation	If cardholder name is more than 26 bytes, the complete cardholder name is placed in this data element.				

Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number				
Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicate Cardholder ID type				
Application expiration date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiration date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in card.				
Application primary account number (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account number on magnetic stripe.				

Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service code defined in magnetic stripe.				
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				
Track 2 equivalent data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				
Cardholder verification method (CVM) list	b	'8E'	14	0000 0000 0000 0000 4203 1E03 1F00 Refer to Section 4.7.12 for details	Lists all CVMs supported by card application in order of priority.. Note: an application can have multiple CVM list, for example: one for domestic transaction and the other for international transaction.				

CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used with RID to identify CA public key in SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N_{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Used for offline data authentication.				
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key exponent	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in SDA process.				

IC card public key certificate	b	'9F46'	N_I	Issuers fill in according to actual situation	IC card public key authenticated by issuer.				
IC card public key exponent	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify signature.				
IC card public key remainder	b	'9F48'	$N_{IC} - N_I + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.				
IC card private key	b	-	N_{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Used for offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.				

Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 4.7.13 for details	Data list that need to be sent to the card from the terminal in internal authentication command; including data object tag and length.				
Application default action (ADA)	b	'9F52'	2	C000 Refer to Section 4.7.14 for details	If issuer authentication is supported, they are UICS specialized data, to define issuer specified actions executed by the card in special conditions. Defaults to all 0 if this data does not exist in card.				
Unique derivation key (UDK) A	b	—	8	Issuers fill in according to actual situation	Unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (UDK) B	b	—	8	Issuers fill in according to actual situation	Unique derivation key of each card is generated by the unique master derivation key of each issuer.				

Unique derivation key (ENC Key) A	b	—	8	Issuers fill in according to actual situation	Enciphered key used for issuer script; unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (ENC Key) B	b	—	8	Issuers fill in according to actual situation	Enciphered key used for issuer script; unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (MAC Key) A	b	—	8	Issuers fill in according to actual situation	Secure message key used for issuer script, unique derivation key of each card is generated by the unique master derivation key of each issuer.				
Unique derivation key (MAC Key) B	b	—	8	Issuers fill in according to actual situation	Secure message key used for issuer script, unique derivation key of each card is generated by the unique master derivation key of each issuer.				

Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.				
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4.7.2 Application Interchange Profile

'7C00' hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

4.7.3 Application Priority Indicator

'01' hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7-5	000	RFU
1	4-1	0001	Top priority

4.7.4 Card Internal Data

Card Internal Data	RFU	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication(DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 = compulsory)	1 byte	00 or 80 Recommend 00	'9F56'
Issuer script command counter	4 bits	00	-
Lower consecutive offline limit	1 byte	Issuers fill in according to actual situation = 0	'9F58'
Upper consecutive offline limit	1 byte	Issuers fill in according to actual situation >= LCOL, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'

Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation =0	'9F54'
Upper cumulative offline transaction limit	6 bytes	Issuers fill in according to actual situation >= CTAL, =0	'9F5C'

Note: To support DDA, SDA, the issuer authentication and authorization control processing, shaded indicator should be set during personalization.

4.7.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

4.7.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E
14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3
Merchant Name And Location	9F4E	20

4.7.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization Response Code	8A	2
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3

4.7.8 Issuer Action Code (Decline, Online and Default)

'00 10 80 00 00' hexadecimal (IAC - decline)

'D8 68 1C F8 00' hexadecimal (IAC - online)

'D8 60 1C A8 00' hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default –Offline decline if cannot go online
Did not perform offline data authentication	Byte 1	0	1	1
Offline static data authentication (SDA) failed		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
Offline dynamic data authentication (DDA) failed		0	1	1
Combined DDA/Application Cryptogram generation (CDA) failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0

RFU		000	000	000
Cardholder verification failed	Byte 3	1	0	0
Unknown CVM		0	0	0
Number of PIN try exceeded the limit		0	0	0
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad exists, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0

Issuer authentication failed		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
Issuer authentication failed		0	0	0
RFU		0000	0000	0000

4.7.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1		Length (15 Max) of self-defined data by Issuer, if any.

4.7.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 03	UnionPay Qcredit

4.7.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back allowed

Byte	B 8	B 7	B 6	B 5	B 4	B 3	B 2	B 1	Usage
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed. If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.7.12 Cardholder Verification Method List

‘0000 0000 0000 0000 4203 1E03 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0100 0010 0000 0011	Online PIN	1	If supported by terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature	2	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.7.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable Number	9F37	4

4.7.14 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform next transaction online.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and the transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	If it is a new card, don't process online transaction.
1	1	0	If it is a new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	0	If PIN is blocked in last transaction, not decline the transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is locked in previous transaction, do not decline transaction and do not block application
2	2	0	Tag 9F79/DF79 NOT reset during GAC processing
2	1	0	RFU

Note: Shaded Indicator Means It Supports Card And Issuer Authentication And Authorization Control Processing.

**4.8 Template 8 — Financial Social Security Card, DDA, Online PIN, Bypass PIN
NOT Allowed, Available Only in Mainland China**

Financial Social Security cards will be only issued in Mainland China, while UICS is for the Issuers outside of Mainland China, thus this template will not be described here.

4.9 Template 9 —Signature Debit Card (Outside of Mainland China Only)— DDA, Signature+Online PIN , Bypass PIN Allowed

4.9.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 4.9.2 for details	Describes the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.9.3 for details	If the card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Version Number	b	'9F08'	2	Initialized. 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its own version No. with that of the card.				
Card Internal Data (The table below summarizes the usage of these internal indicators.)	var.	-		Refer to Section 4.9.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence the transaction results.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates that the last online transaction hasn't been finished.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Online request by card indicator	b	-	1 bit	The initial setting is 0	During the transaction processing, set when the card decides transaction online.				
Offline decline requested by Card indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates the issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Labels whether SDA fails, when the last online transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of the last submission of online transaction				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records. <i>UICS</i> has provided the recommended values: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.9.5 for details	List data object tag and length in log record				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommend value 00	A maximum number of consecutive offline transactions not using specified application currency. Requests online transaction when this number is exceeded				
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommend value 00	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests online transaction when the counter exceeds the cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00 00 00 00 00 00	Records the cumulative offline transaction amount since the completion of last online transaction using currency specified by the application				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommend value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. The transaction requests to go online when this number is exceeded				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommend value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If exceeded and the transaction cannot go online, the transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommend value 00	Maximum consecutive offline transaction limit allowed by the card before applying for online authorization.				
Upper consecutive offline transaction limit(UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommend value 00	Upper consecutive offline transaction limit allowed by the card before required by the issuer; when this value is exceeded, the transaction will be declined if the request to go online fails.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.1.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), the data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to Section 4.9.7 for details	Lists the data that the card requests the terminal to transmit of in second Generate AC. Its content is terminal data objects (tag and length), the data include: the issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time				
Cryptogram information data	b	'9F27'	1	The initial setting is 0	Indicates the type of cryptogram returned by the card				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer Action Code (IAC) - decline	b	'9F0E'	5	00 10 80 00 00 Refer to Section 4.9.8 for details	Specifies the condition that cause the transaction to be declined directly without going online.				
Issuer Action Code (IAC) - online	b	'9F0F'	5	D8 68 1C F8 00 Refer to 4.9.8 for details	Specifies the conditions for transaction online submission.				
Issuer Action Code (IAC) - default	b	'9F0D'	5	D0 60 1C A8 00 Refer to 4.9.8 for details	Specify conditions for transaction rejection when transaction request to go online but the terminal cannot finish online submission.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer application data	b	'9F10'	Var.	07_ _ 01 03 00 00 00 01Refer to Section 4.9.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.				
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Specifies the country of card issuer. Please refer to Appendix A.3				
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to the issuer preferred language.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application currency code	n	'9F51'	2	Issuers fill in according to actual situation	Domestic currency of the Issuer. Please refer to Appendix A.3				
Application Identifier (AID)	b	'4F'	5-16	Initialized Refer to Section 4.9.10 for details	UnionPay debit card AID: A0 00 00 03 33 01 01 01				
Application label	ans	'50'	1-16	Issuers fill in according to actual situation Refer to Section 4.9.10 for details	Application name when the terminal displays an optional application list to the consumer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Usage Control	b	'9F07'	2	FF 00 Refer to Section 4.9.11 for details	Labels certain card-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).				
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to specify different applications using the same account in the card.				
Cardholder name	ans	'5F20'	2-26	Provided in magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder name extension	ans	'9F0B'	27—45	Issuers fill in according to actual situation	If cardholder name exceeds 26 bytes, the complete part is placed in this data element.				
Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID No.				
Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicates cardholder ID type				
Application expiry date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiry date				
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application primary account No. (PAN)	cn	'5A'	10 at maximum	Provided by magnetic stripe data files	Equivalent to application primary account No. in magnetic stripe.				
Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service codes defined in magnetic stripe.				
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.				
Track 2 equivalent data	var.	'57'	19 at maximum	Provided by magnetic stripe data files	Equivalent to track 2 data (but the format is not same) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder Verification Method (CVM) list	b	'8E'	14	0000 0000 0000 0000 1E03 0203 1F00 Refer to Section 4.9.12 for details	Lists all CVMs supported by card application in order of priority. Note: An application can have multiple CVM lists, for example: one is for domestic transaction and the other is for cross-border transaction.				
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used together with RID to identify CA public key in the SDA or DDA process.				
Issuer public key (IPK) certificate	b	'90'	N _{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.				
Issuer public key index	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.				
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, and verified by the terminal in the SDA process.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Default Action (ADA)	b	'9F52'	2	C000 Refer to Section 4.9.13 for details	If issuer authentication is supported, They are UICS specialized data defining issuer specified actions carried out by the card under special conditions. All defaults set as 0 if this data does not exist in the card.				
Unique DEA Key (UDK) A	b	—	8	Issuers fill in according to actual situation	The Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (UDK) B	b	—	8	Issuers fill in according to actual situation	The Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (ENC Key)A	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key)B	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key)A	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (MAC Key)B	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank ac-count. The same PAN and its tokens have one identical PAR.				

4.9.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

4.9.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Without cardholder confirmation, application can be selected.
1	7-5	000	RFU
1	4-1	0001	Top priority

4.9.4 Card Internal Data

Card Internal Data	Reserved	Initial Value	Tag
Online authorization indicator	1 bit	0	-
Card request ARQC indicator	1 bit	0	-

Card Internal Data	Reserved	Initial Value	Tag
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit		-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 = mandatory)	1 byte	00 or 80, Recommend 00	'9F56'
Issuer script command counter	4 bits		-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation =0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation $\geq$ lower consecutive offline transaction limit, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'

Card Internal Data	Reserved	Initial Value	Tag
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation=0	'9F54'
Upper cumulative offline transaction limit	6 bytes	>=Cumulative offline transaction amount limit (domestic), =0	'9F5C'

Note: To support SDA, issuer authentication and authorization control processing, indicator in the shadow shall be set during the personalization process.

4.9.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction Date	9A	3
Transaction Time	9F21	3
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Transaction Currency Code	5F2A	2

Data Object Name	Tag	Length
Merchant Name and Location	9F4E	20
Transaction Type	9C	1
Application Transaction Counter (ATC)	9F36	2

4.9.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3
Merchant Name and Location	9F4E	20

4.9.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization Response Code	8A	2
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3

4.9.8 Issuer Action Code (Decline, Online and Default)

'00 10 80 00 00' hexadecimal (IAC - decline)

'D8 68 1C F8 00' hexadecimal (IAC - online)

'D0 60 1C A8 00' hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline - Offline decline	IAC online – Request online	IAC default – Offline reject if cannot go online
Offline data authentication not performed	Byte 1	0	1	1
SDA fails		0	1	1
IC card data missing		0	0	0
Card appears in terminal exception file		0	1	1
DDA fails		0	1	0
CDA failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification fails	Byte 3	1	0	0
Unknown CVM		0	0	0

Number of PIN try exceeds the limit		0	0	0
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad present, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds the lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication fails		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.9.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm indicator
9	8-1		If issuer defined data present, self-defined data length (15 max)

4.9.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 01	UnionPay Debit

4.9.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed. If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.9.12 Cardholder Verification Method List

‘0000 0000 0000 0000 1E03 0203 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0001 1110 0000 0011	Signature	1	If supported by terminal	Fail Cardholder Verification
0000 0010 0000 0011	Online PIN	2	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.9.13 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform transaction online next time.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	In case of a new card, don't process online transaction.
1	1	0	In case of new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.

Byte	Bit	Value	Definition
2	7	0	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is blocked in previous transaction, do not decline transaction or block application
2	2	0	Tag '9F79'/ 'DF79' NOT reset during GAC processing
2	1	0	RFU

Note: Indicator in the shadow represents that it supports card and issuer authentication and authorization control processing.

4.10 Template 10 — Debit Card (U.S. Only) — DDA, Online PIN+Signature, Bypass PIN Allowed

4.10.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7C 00 Refer to Section 4.10.2 for details	Describes the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 4.10.3 for details	If the card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Version Number	b	'9F08'	2	Initialized. 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its own version No. with that of the card.				
Card Internal Data (The table below summarizes the usage of these internal indicators.)	var.	-		Refer to Section 4.10.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence the transaction results.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates that the last online transaction hasn't been finished.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Online request by card indicator	b	-	1 bit	The initial setting is 0	During the transaction processing, set when the card decides transaction online.				
Offline decline requested by Card indicator	b	-	1 bit	The initial setting is 0	During transaction processing, set when the card decides to decline the transaction.				
Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates the issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Labels whether SDA fails, when the last online transaction is declined.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction going online. Issuer authentication can be optional ('00') or mandatory ('80'). If it is mandatory but authorization response cryptogram is not returned, the Issuer can choose to decline this transaction irrespective of the result of online response message.				
Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of the last submission of online transaction				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records. <i>UICS</i> has provided the recommended values: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 4.10.5 for details	List data object tag and length in log record	☐			
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommend value 00	A maximum number of consecutive offline transactions not using specified application currency. Requests online transaction when this number is exceeded		☐		

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommend value 00	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded		☐		
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests online transaction when the counter exceeds the cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00 00 00 00 00 00	Records the cumulative offline transaction amount since the completion of last online transaction using currency specified by the application				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommend value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. The transaction requests to go online when this number is exceeded		☐		
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommend value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If exceeded and the transaction cannot go online, the transaction is declined.		☐		

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommend value 00	Maximum consecutive offline transaction limit allowed by the card before applying for online authorization.		☐		
Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommend value 00	Upper consecutive offline transaction limit allowed by the card before required by the issuer; when this value is exceeded, the transaction will be declined if the request to go online fails.		☐		

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 4.10.6 for details	List the data that the card requests the terminal to transmit in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), the data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location	☐			☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Card Risk Management Data Object List2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to Section 4.10.7 for details	Lists the data that the card requests the terminal to transmit of in second Generate AC. Its content is terminal data objects (tag and length), the data include: the issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time	☐			☐
Cryptogram information data	b	'9F27'	1	The initial setting is 0	Indicates the type of cryptogram returned by the card	☐			

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer Action Code (IAC) - decline	b	'9F0E'	5	00 10 80 00 00 Refer to Section 4.10.8 for details	Specifies the condition that cause the transaction to be declined directly without going online.	☐			☐
Issuer Action Code (IAC) - online	b	'9F0F'	5	D8 68 1C F8 00 Refer to 4.10.8 for details	Specifies the conditions for transaction online submission.	☐			☐
Issuer Action Code (IAC) - default	b	'9F0D'	5	D0 60 1C A8 00 Refer to 4.10.8 for details	Specify conditions for transaction rejection when transaction request to go online but the terminal cannot finish online submission.	☐			☐
Issuer application data	b	'9F10'	Var.	07 _ _ 01 03 00 00 00 01 Refer to Section 4.10.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.		☐		

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicates the country where the card issuer is located Please refer to Appendix A.3		☐		☐
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to the issuer preferred language.		☐		
Application currency code	n	'9F51'	2	Issuers fill in according to actual situation	Domestic currency of the Issuer. Please refer to Appendix A.3		☐		☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Identifier (AID)	b	'4F'	5-16	Initialized Refer to Section 4.10.10 for details	UnionPay debit card AID: A0 00 00 03 33 01 01 01 UnionPay debit cards and some prepaid card issued in the U.S. shall support the U.S. Common Debit AID: A0 00 00 03 33 01 01 08 <i>Note: If the U.S. Common Debit AID is supported, the two AIDs above shall point to the same debit application.</i>				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Label	ans	'50'	1-16	Fill in by Issuers according to the actual situation Refer to Section 4.10.10 for details	Application name when the terminal displays an optional application list to the consumer.		☐		
Application Usage Control	b	'9F07'	2	FF 00 Refer to Section 4.10.11 for details	Labels certain card-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).	☐			☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Primary Account Sequence Number	n	'5F34'	1	Issuers fill in according to actual situation	Used to specify different applications using the same account in the card.			☐	☐
Cardholder Name	ans	'5F20'	2-26	Provided in magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element. Should not write the real full name of cardholder into the application			☐	☐
Cardholder Name Extension	ans	'9F0B'	27—45	Issuers fill in according to actual situation	If cardholder name exceeds 26 bytes, the complete part is placed in this data element. Should not write the real full name of cardholder into the application			☐	☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder ID Number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID No. Should not write the real full ID No. of cardholder into the application			☐	☐
Cardholder ID Type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicates cardholder ID type			☐	☐
Application Expiry Date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiry date			☐	☐
Application Effective Date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in the card.		☐		☐
Application Primary Account No. (PAN)	cn	'5A'	10 at maximum	Provided by magnetic stripe data files	Equivalent to application primary account No. in magnetic stripe.			☐	☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Service Code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service codes defined in magnetic stripe.			☐	☐
Track 1 Self-Defined Data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.			☐	☐
Track 2 Equivalent Data	var.	'57'	19 at maximum	Provided by magnetic stripe data files	Equivalent to track 2 data (but the format is not same) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card			☐	☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder Verification Method (CVM) list	b	'8E'	14	0000 0000 0000 0000 4203 1E03 1F00 Refer to Section 4.10.12 for details	Lists all CVMs supported by the card application in order of priority. <i>Note: An application can have multiple CVM lists, for example: one is for domestic transactions and the other is for cross-border transactions.</i>	☐			☐
CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used together with RID to identify CA public key in the SDA or DDA process.		☐		☐
Issuer public key (IPK) certificate	b	'90'	N _{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Applied to offline data authentication.		☐		☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.		☐		☐
Issuer public key index	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.		☐		☐
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, and verified by the terminal in the SDA process.			☐	☐

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Default Action (ADA)	b	'9F52'	2	C000 Refer to Section 4.10.13 for details	If issuer authentication is supported, They are UICS specialized data defining issuer specified actions carried out by the card under special conditions. All defaults set as 0 if this data does not exist in the card.	☐			
Unique DEA Key (UDK) A	b	—	8	Issuers fill in according to actual situation	The Unique DEA Key of each card is generated by the unique master derivation key of each issuer.			☐	
Unique DEA Key (UDK) B	b	—	8	Issuers fill in according to actual situation	The Unique DEA Key of each card is generated by the unique master derivation key of each issuer.			☐	

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (ENC Key)A	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (ENC Key)B	b	—	8	Issuers fill in according to actual situation	Enciphered key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Unique DEA Key (MAC Key)A	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Unique DEA Key (MAC Key)B	b	—	8	Issuers fill in according to actual situation	Secure message key applied to issuer script; the Unique DEA Key of each card is generated by the unique master derivation key of each issuer.				
Payment Account Reference (PAR)	an	'9F24'	29	If card uses Token, issuer should fill it in.	Payment Account Reference (PAR) is a data element that is used to associate different tokens deriving from the same bank account. The same PAN and its tokens have one identical PAR.			☐	

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Issuer Identification Number (IIN)	n	'42'	3		The number that identifies the major industry and the card issuer and that forms the first part of the Primary Account Number (PAN). In template 'BF0C' or '73'.				

4.10.2 Application Interchange Profile (AIP)*‘7C00’ hexadecimal*

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

4.10.3 Application Priority Indicator*‘01’ hexadecimal*

Byte	Bit	Value	Definition
1	8	0	Without cardholder confirmation, application can be selected.
1	7-5	000	RFU
1	4-1	0001	Top priority

4.10.4 Card Internal Data

Card Internal Data	Reserved	Initial Value	Tag
Online authorization indicator	1 bit	0	-

Card Internal Data	Reserved	Initial Value	Tag
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication (DDA) failure indicator	1 bit		-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 = mandatory)	1 byte	00 or 80, Recommended 00	'9F56'
Issuer script command counter	4 bits		-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation =0	'9F58'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation >=lower consecutive offline transaction limit, =0	'9F59'
Last online application transaction counter (ATC) register	2 bytes	0	'9F13'

Card Internal Data	Reserved	Initial Value	Tag
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F53'
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation=0	'9F54'
Cumulative Total Transaction Amount Upper Limit	6 bytes	>=Cumulative total transaction amount limit (domestic), =0	'9F5C'

Note: To support SDA, issuer authentication and authorization control processing, indicator in the shadow shall be set during the personalization process.

4.10.5 Log Format

‘9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction Date	9A	3
Transaction Time	9F21	3
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Transaction Currency Code	5F2A	2
Merchant Name and Location	9F4E	20
Transaction Type	9C	1
Application Transaction Counter (ATC)	9F36	2

4.10.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5

Data Object Name	Tag	Length
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3
Merchant Name and Location	9F4E	20

4.10.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization Response Code	8A	2
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
Transaction Time	9F21	3

4.10.8 Issuer Action Code (Decline, Online and Default)

'00 10 80 00 00' hexadecimal (IAC - decline)

'D8 68 1C F8 00' hexadecimal (IAC - online)

'D0 60 1C A8 00' hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline - Offline decline	IAC online – Request online	IAC default – Offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA fails		0	1	1
IC card data missing		0	0	0
Card appears in terminal exception file		0	1	1
DDA fails		0	1	0
CDA failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification fails	Byte 3	1	0	0

Part I Personalization Template Based on Debit/Credit Application

Unknown CVM		0	0	0
Number of PIN try exceeds the limit		0	0	0
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad present, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds the lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication fails		0	0	0

Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

4.10.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm indicator
9	8-1		If issuer defined data present, self-defined data length (15 max)

4.10.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 01	UnionPay Debit

Application Identifier (AID)		Application Label
	01 01 08	U.S. Common Debit <i>Note: The Application Label of the U.S. Common Debit AID must not contain "UnionPay".</i>

4.10.11 Application Usage Control

‘FF00’ hexadecimal

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back
2	0	0	0	0	0	0	0	0	International cash back allowed

Byte	B8	B7	B6	B5	B4	B3	B2	B1	Usage
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed .If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

4.10.12 Cardholder Verification Method List

‘0000 0000 0000 0000 4203 1E03 1F00’ hexadecimal

CVM Code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0100 0010 0000 0011	Online PIN	1	If supported by the terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature	2	If supported by the terminal	Cardholder verification fails
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

4.10.13 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
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Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform transaction online next time.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	In case of a new card, don't process online transaction.
1	1	0	In case of new card, do not decline transaction when transaction cannot go online.
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	0	If PIN is blocked in last transaction, do not decline transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is blocked in previous transaction, do not decline transaction or block application

Byte	Bit	Value	Definition
2	2	0	Reset Tag 9F79/DF79 during GAC processing
2	1	0	RFU

Note: Indicator in the shadow represents that it supports card and issuer authentication and authorization control processing.

5 Low-value Payment and qUICS Personalization Templates

The personalized template for e-cash feature includes seven templates, of which Template 11 is a pure e-cash one. The e-cash template is a template for the type of unregistered prepaid cards. The template creates small-amount accounts but not accounts of auto-binding debt cards or credit cards. Therefore, it includes standard data template for debit and credit, small-amount payment template based on debit and credit and small-amount payment template for quick debit and credit, and can be used independently. The other four e-cash templates are functional templates. When issuers use any of them, they should first select a standard debit/credit template and then the e-cash template for the use. For particulars, please refer to Schedule A.2.3.

5.1 Template 11 — Pure Electronic Cash Card

5.1.1 Standard UICS Debit/Credit Data Template — DDA, Online PIN, Bypass PIN NOT Allowed

5.1.1.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 5.1.1.2 for details	Indicates the functions supported by the card in this application.				
Application Priority Indicator	b	'87'	1	01 Refer to Section 5.1.1.3 for details	If card has multiple applications, indicates application priority in the same directory.				
Application Transaction Counter (ATC)	b	'9F36'	2	The initial setting is 00 00	Records number of transaction processed after personalization.				

Application Version Number	b	'9F08'	2	Initialized 00 30	Version No. assigned to the application by payment system, providing ways to add new functions in the future. The terminal will compare its version No. with that of the card.				
Card Internal Data (next table summarizes usage of these internal indicators)	var.	-		Refer to Section 5.1.1.4 for details	Card internal counter and indicator used by the Issuer to provide transaction processing information and influence transaction result.				
Last online transaction incomplete indicator	b	-	1 bit	The initial setting is 0	Indicates last online transaction incomplete.				
Online request by card indicator	b	-	1 bit	The initial setting is 0	In transaction processing, set when card decides transaction online.				
Offline decline requested by card indicator	b	-	1 bit	The initial setting is 0	In transaction processing, set when the card decides to decline the transaction.				

Issuer authentication failure indicator	b	-	1 bit	The initial setting is 0	Indicates that issuer authentication error occurred in last transaction.				
Static data authentication (SDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether SDA failed when the last online transaction was declined.				
Dynamic data authentication (DDA) failure indicator	b	-	1 bit	The initial setting is 0	Indicates whether DDA failed when the last online transaction was declined.				
Issuer authentication indicator	b	'9F56'	1	00 or 80 Recommend 00	Indicator to control transaction processing after transaction online. Issuer authentication can be optional ('00') or compulsory ('80'). If it is compulsory but authorization response cryptogram is not returned, the Issuer can choose to reject this transaction irrespective the result of online response message.				

Last online application transaction counter (ATC) register	b	'9F13'	2	The initial setting is 00 00	ATC value of last online transaction				
Log entry	b	'9F4D'	2	0B 0A	Provides SFI of log files and number of log file records, <i>UICS Specification</i> has provided recommendation value: 0B 0A Byte 1: SFI of circular transaction log file is 11 (decimal) Byte 2: Number of records in transaction log file is 10 (decimal)				

Log format	b	'9F4F'	Var.	9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02 Refer to Section 5.1.1.5 for details	Lists data object tag and length in log records				
Consecutive offline transaction limit (international - currency)	b	'9F53'	1	Issuers fill in according to actual situation Recommended value 00	Maximum number of consecutive offline transactions not using specified application currency, the transaction requests to go online when this number is exceeded				

Consecutive offline transaction limit (international - country)	b	'9F72'	1	Issuers fill in according to actual situation Recommended value 00	Maximum number of consecutive offline transactions not in Issuer country, the transaction requests to go online when this number is exceeded				
Cumulative transaction counter (international - currency)	b	-	1	The initial setting is 00	International offline transaction counters. The card requests transaction for online processing when counter exceeds cumulative offline transaction limit.				
Cumulative offline transaction amount (domestic)	n	-	6	The initial setting is 00	Records offline transaction cumulated amount since the completion of last online transaction using currency specified by the application				

Cumulative total transaction amount limit	n	'9F54'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount. Request online for transaction exceeding this limit				
Cumulative Total Transaction Amount Upper Limit	n	'9F5C'	6	Issuers fill in according to actual situation Recommended value 00 00 00 00 00 00	Maximum limit of cumulative offline transaction amount and cumulative offline transaction amount (dual currency). If it is exceeded and the transaction cannot go online, the transaction is declined.				
Lower consecutive offline transaction limit (LCOL)	b	'9F58'	1	Issuers fill in according to actual situation Recommended value 00	Maximum consecutive offline transaction limit allowed by card before applying for online authorization.				

Upper consecutive offline transaction limit (UCOL)	b	'9F59'	1	Issuers fill in according to actual situation Recommended value 00	Upper consecutive offline limit allowed by the card before requesting online processing, which is specified by the Issuer; when this value is exceeded, the transaction will be declined if the request to go online failed.				
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Card Risk Management Data Object List 1 (CDOL1)	b	'8C'	27	9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14 Refer to Section 5.1.1.6 for details	List the data that the card requests the terminal to transmit l in the first command for generating application cryptogram. It is used to support cryptogram version 01 and authorization control processing. Its content is terminal data objects (tag and length), data include: authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, transaction time, and Merchant name and location.				
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Card Risk Management Data Object List 2 (CDOL2)	b	'8D'	26	8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 Refer to Section 5.1.1.7 for details	List the data that the card requests the terminal to transmit in second Generate AC. Its content is terminal data objects (tag and length), data include: issuer response code, authorized amount, other amounts, terminal country code, terminal verification result, transaction currency code, transaction date, transaction type, terminal unpredictable number, and transaction time.					
Cryptogram information data	b	'9F27'	1	The initial setting is 00	Indicate the type of cryptogram returned by card					
Issuer action code (IAC) - decline	b	'9F0E'	5	00 10 98 00 00 Refer to Section 5.1.1.8 for details	Specifies the conditions for declining the transaction directly without going online.					

Issuer action code (IAC) - online	b	'9F0F'	5	D8 68 04 F8 00 Refer to Section 5.1.1.8 for details	Specifies the conditions for transaction online submission.					
Issuer action code (IAC) - default	b	'9F0D'	5	D8 60 04 A8 00 Refer to Section 5.1.1.8 for details	Specifies Conditions for transaction declining when the transaction request online but terminal cannot complete online escalation.					
Issuer application data	b	'9F10'	Var.	Refer to Section 5.1.1.9 for details	Specific application data that need to be sent to the Issuer in an online transaction.					
Issuer country code	n	'5F28'	2	Issuers fill in according to actual situation	Indicate the country of card issuer. Please refer to Appendix A.3					
Preferred language	an	'5F2D'	2	Issuers fill in according to actual situation	When the terminal supports multiple languages, the terminal displays information according to issuer preferred language.					

Application currency code	n	'9F51'		Issuers fill in according to actual situation	Issuer domestic currency. Please refer to Appendix A.3					
Application Identifier(AID)	b	'4F'	5-16	Initialized Refer to Section 5.1.1.10 for details	Registered Application Provider Identifier (RID) and proprietary identifier extension: A0 00 00 03 33 01 01 06 A0 00 00 03 33 determines UICS registered application provider (same for all cards), 01 01 06 indicates UICS e-cash application.					
Application Label	ans	'50'	1-16	Fill in by Issuers according to the actual situation Refer to Section 5.1.1.10 for details	Application name when the terminal displays an optional application list to the consumer.					

Application Usage Control	b	'9F07'	2	FF 00 Refer to Section 5.1.1.11 for details	Indicates certain issuer-specific restrictions on card application, including geographic usage and service type, etc. Used to provide more flexible card service control (similar to service code).					
Application primary account sequence number	n	'5F34'	1	Issuers fill in according to actual situation	Used to indicate different applications using the same account in the card.					
Cardholder name	ans	'5F20'	2-26	Provided by magnetic stripe data files	Equivalent to cardholder name in magnetic stripe. If cardholder name is no more than 26 bytes, the complete cardholder name is placed in this data element.					
Cardholder name extension	ans	'9F0B'	27 - 45	Issuers fill in according to actual situation	If cardholder name is more than 26 bytes, the complete cardholder name is placed in this data element.					

Cardholder ID number	an	'9F61'	1-40	Issuers fill in according to actual situation	Cardholder ID number					
Cardholder ID type	cn	'9F62'	1	Issuers fill in according to actual situation	Indicate Cardholder ID type					
Application expiration date	n	'5F24'	3	Provided by magnetic stripe data files	Card application expiration date					
Application effective date	n	'5F25'	3	Issuers fill in according to actual situation	Application initiation date in card.					
Application primary account number (PAN)	cn	'5A'	10 max	Provided by magnetic stripe data files	Equivalent to application primary account number on magnetic stripe.					

Service code	n	'5F30'	2	Provided by magnetic stripe data files	The same as service code defined in magnetic stripe.					
Track 1 self-defined data	ans	'9F1F'	var.	Provided by magnetic stripe data files	The same as track 1 self-defined data in magnetic stripe.					
Track 2 equivalent data	var.	'57'	Maximum of 19	Provided by magnetic stripe data files	Equivalent to track 2 data (format is not equivalent) Effective date of track 2 equivalent data must match the application expiration date (5F24) in the IC card					
Cardholder verification method (CVM) list	b	'8E'	12	0000 0000 0000 0000 0203 1F00 Refer to Section 5.1.1.12 for details	Lists all CVMs supported by card application in order of priority.e. Note: an application can have multiple CVM list, for example: one for domestic transaction and the other for international transaction.					

CA public key index (PKI)	b	'8F'	1	Issuers fill in according to actual situation	Used with RID to identify CA public key in SDA or DDA process.					
Issuer public key (IPK) certificate	b	'90'	N_{CA}	Issuers fill in according to actual situation	Issuer public key authenticated by CA. Used for offline data authentication.					
Issuer public key remainder (If required)	b	'92'	$N_1 - N_{CA} + 36$	Issuers fill in according to actual situation	Issuer public key part not placed in the issuer public key certificate.					
Issuer public key index	b	'9F32'	1 to $N_1/4$	Issuers fill in according to actual situation	Issuer public key index, used to verify signed static application data and IC card public key certificate.					
Signed static application data (SAD)	b	'93'	var.	Issuers fill in according to actual situation	Application data signed by issuer private key, verified by the terminal in SDA process.					

IC card public key certificate	b	'9F46'	N_I	Issuers fill in according to actual situation	IC card public key authenticated by issuer.					
IC card public key index	b	'9F47'	1 or 3	Issuers fill in according to actual situation	Dynamic application data used by IC card public key index to verify signature.					
IC card public key remainder	b	'9F48'	$N_{IC} - N_I + 42$	Issuers fill in according to actual situation	IC card public key part not placed in the IC card public key certificate.					
IC card private key	b	-	N_{IC}	Issuers fill in according to actual situation	Private key part of IC card public key pair. Used for offline dynamic data authentication. There are two formats: module/private key index form, and Chinese Remainder Theorem (CRT) form.					

Dynamic data authentication data object list (DDOL)	b	'9F49'	252 max	Issuers fill in according to actual situation Refer to Section 5.1.1.10 for details	Data list that need to be sent to the card from the terminal in internal authentication command; including data object tag and length.					
Application default action (ADA)	b	'9F52'	2	C000 Refer to Section 5.1.1.14 for details	If issuer authentication is supported, they are UICS specialized data, to define issuer specified actions executed by the card in special conditions. Defaults to all 0 if this data does not exist in card.					
Unique derivation key (UDK) A	b	—	8	Issuers fill in according to actual situation	Unique derivation key of each card is generated by the unique master derivation key of each issuer.					

Unique derivation key (UDK) B	b	—	8	Issuers fill in according to actual situation	Unique derivation key of each card is generated by the unique master derivation key of each issuer.					
Unique derivation key (ENC Key) A	b	—	8	Issuers fill in according to actual situation	Enciphered key used for issuer script; unique derivation key of each card is generated by the unique master derivation key of each issuer.					
Unique derivation key (ENC Key) B	b	—	8	Issuers fill in according to actual situation	Enciphered key used for issuer script; unique derivation key of each card is generated by the unique master derivation key of each issuer.					
Unique derivation key (MAC Key) A	b	—	8	Issuers fill in according to actual situation	Secure message key used for issuer script, unique derivation key of each card is generated by the unique master derivation key of each issuer.					

Unique derivation key (MAC Key) B	b	—	8	Issuers fill in according to actual situation	Secure message key used for issuer script, unique derivation key of each card is generated by the unique master derivation key of each issuer.					
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5.1.1.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

5.1.1.3 Application Priority Indicator

‘01’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Application can be selected without cardholder confirmation.
1	7-5	000	RFU
1	4-1	0001	Top priority

5.1.1.4 Card Internal Data

Card Internal Data	RFU	Initial Value	Tag
Online authorization indicator	1 bit	0	-

Card Internal Data	RFU	Initial Value	Tag
Card request ARQC indicator	1 bit	0	-
Card request AAC indicator	1 bit	0	-
Issuer authentication failure indicator	1 bit	0	-
Static data authentication (SDA) failure indicator	1 bit	0	-
Dynamic data authentication(DDA) failure indicator	1 bit	0	-
Issuer script failure indicator	1 bit		-
Issuer authentication indicator (bit8 0 = optional 1 =mandatory)	1 byte	00 or 80 Recommen d 00	'9F5 6'
Issuer script command counter	4 bits		-
Lower consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation = 0	'9F5 8'
Upper consecutive offline transaction limit	1 byte	Issuers fill in according to actual situation > = LCOL, =0	'9F5 9'
Last online application transaction counter (ATC) register	2 bytes	0	'9F1 3'
Consecutive offline transaction limit (international - currency)	1 byte	Issuers fill in according to actual situation =0	'9F5 3'

Card Internal Data	RFU	Initial Value	Tag
Consecutive offline transaction limit (international - country)	1 byte	Issuers fill in according to actual situation =0	'9F72'
Cumulative offline transaction counter (international)	1 byte	0	-
PIN try limit	1 byte	Issuers fill in according to actual situation	-
PIN try number counter	1 byte	= PIN try limit	'9F17'
Cumulative offline transaction amount (domestic)	6 bytes	0	-
Cumulative total transaction amount limit (domestic)	6 bytes	Issuers fill in according to actual situation = 0	'9F54'
Cumulative Total Transaction Amount Upper Limit	6 bytes	Issuers fill in according to actual situation > = CTAL, =0	'9F5C'

Note: To support DDA, SDA, the issuer authentication and authorization control processing, shaded indicator should be set during personalization.

5.1.1.5 Log Format

'9A 03 9F21 03 9F02 06 9F03 06 9F1A 02 5F2A 02 9F4E 14 9C 01 9F36 02'
hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Transaction currency code	5F2A	2
Merchant name and location	9F4E	20
Transaction type	9C	1
Application transaction counter (ATC)	9F36	2

5.1.1.6 Card Risk Management Data Object List 1

‘9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1

Data Object Name	Tag	Length
Unpredictable number	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	20

5.1.1.7 Card Risk Management Data Object List 2

‘8A 02 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Authorization response code	8A	2
Authorized amount	9F02	6
Other amounts	9F03	6
Terminal country code	9F1A	2
Terminal verification result	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3

5.1.1.8 Issuer Action Code (Decline, Online and Default)

‘00 10 98 00 00’ hexadecimal (IAC - decline)

‘D8 68 04 F8 00’ hexadecimal (IAC - online)

‘D8 60 04 A8 00’ hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA failed		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
DDA failed		0	1	1
CDA failed		0	0	0
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
CVM fails	Byte 3	1	0	0
Unknown CVM		0	0	0

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
Times of PIN try exceeds the limit		0	0	0
PIN entry required, but PIN pad not present or not working		1	0	0
PIN entry required, PIN pad exists, but no PIN is entered		1	0	0
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
Use default TDOL	Byte 5	0	0	0
Issuer authentication failed		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

5.1.1.9 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1	0A	If issuer self-defined data exist, self-defined data length (15 max)

Byte	Bit	Hexadecimal Initial Value	Conditions
10	8-1	01	IDD ID
11-15	8-1	Initialized as 0	Field indicating amount
16-19	8-1	Initialized as 0	MAC

5.1.1.10 Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	proprietary extension(PIX) identifier	Card Type
A0 00 00 03 33	01 01 06	Self-defined based on debit/credit product

5.1.1.11 Application Usage Control

‘FF00’ hexadecimal

Byte	b8	b7	b6	b5	b4	b3	b2	B1	Usage
1	1	0	0	0	0	0	0	0	Valid for domestic cash transaction
1	0	1	0	0	0	0	0	0	Valid for international cash transaction
1	0	0	1	0	0	0	0	0	Valid for domestic product
1	0	0	0	1	0	0	0	0	Valid for international product
1	0	0	0	0	1	0	0	0	Valid for domestic service
1	0	0	0	0	0	1	0	0	Valid for international service

B y t e	b 8	b 7	b 6	b 5	b 4	b 3	b 2	B 1	Usage
1	0	0	0	0	0	0	1	0	Valid for ATM
1	0	0	0	0	0	0	0	1	Valid for terminals except ATM
2	0	0	0	0	0	0	0	0	Domestic cash back allowed
2	0	0	0	0	0	0	0	0	International cash back allowed
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU
2	0	0	0	0	0	0	0	0	RFU

Note: This example shows the settings if the Service Code on the magnetic stripe is set to 2. If bit is set to 1 the service is allowed. If the Service Code on the magnetic stripe is set to 6, only domestic functions are permitted.

5.1.1.12 Cardholder Verification Method List

‘0000 0000 0000 0000 0203 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Conditions	Action if CVM Fails
0000 0010 0000 0011	Online PIN	1	If supported by the terminal	Fail Cardholder Verification

CVM code – first eight bytes are 0	CVM	Processing Order	Conditions	Action if CVM Fails
0001 1111 0000 0000	CVM not required	2	Always	Will not fail

5.1.1.13 Dynamic Data Object List

‘9F37 04’ hexadecimal

(Data object tag and length)

Value	Tag	Length
Unpredictable number	9F37	4

5.1.1.14 Application Default Action

‘C000’ hexadecimal

Byte	Bit	Value	Definition
1	8	1	If issuer authentication fails, perform next transaction online.
1	7	1	If issuer authentication executed but failed, decline the transaction.
1	6	0	If issuer authentication is necessary but ARPC not received, do not decline the transaction.
1	5	0	If transaction is declined, do not generate notification.
1	4	0	If PIN is blocked in this transaction and the transaction is declined, do not generate notification.
1	3	0	If transaction is declined because issuer authentication fails or is not performed, do not generate notification.
1	2	0	If it is a new card, don't process online transaction.
1	1	0	If it is a new card, do not decline transaction when transaction cannot go online.

Byte	Bit	Value	Definition
2	8	0	If PIN is blocked in this transaction, do not block application.
2	7	0	If PIN is blocked in last transaction, not decline the transaction.
2	6	0	If PIN is blocked in last transaction, don't process online transaction.
2	5	0	If PIN is blocked in last transaction, do not decline transaction when transaction cannot go online
2	4	0	If issuer script command failed in last transaction, don't process online transaction.
2	3	0	If PIN is locked in previous transaction, do not decline transaction and do not block application
2	2	0	Reset Tag 9F79/DF79 during GAC processing
2	1	0	RFU

Note: Indicator in shadow represent it supports card and issuer authentication and authorization control processing.

5.1.2 Low-value Payment Based on Debit/Credit Application – CDA, CVM NOT Required, E-cash Load Logs

5.1.2.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Application Interchange Profile (AIP)	b	'82'	2	7D00 Refer to Section 5.1.2.2 for details	Indicate the functions supported by Low-value payment for Debit/Credit				
Electronic Cash Balance	n	'9F79'	6	The initial setting is 0	Keeps the remaining amount for offline purchase transactions.				
Electronic Cash Balance Upper Limit	n	'9F77'	6	Issuers fill in according to actual situation	Represents the max cumulative amount the cardholder can spend on offline purchases in the electronic cash application, and it is the upper limit for topping up the card.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Electronic Cash Issuer Authorization Code	an	'9F74'	6	ECC001	The Code on the card, indicating the approval of electronic cash transactions.				
Transaction Amount Limit	n	'9F78'	6	Issuers fill in according to actual situation	Upper limit of single transaction amount specified by the card, to control risk of single electronic cash transaction.				
Electronic Cash Reset Amount	n	'9F6D'	6	Issuers fill in according to actual situation	Lower limit of available balance, to initiate card automatic loading.				
Processing Options Data Objects (PDOL)	b	'9F38'	9	9F7A 01 9F02 06 5F2A 02 Please refer to Section 5.1.2.3 for details.	The data sent by the terminal to the card, included in GET PROCESSING OPTIONS command, containing terminal data objects (tags and lengths)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Cardholder Verification Method (CVM) List	b	'8E'	10	0000 0000 0000 0000 1F03 Please refer to Section 5.1.2.4 for details.	Lists all CVMs supported by the card in order of priority.. Note: there can be more than one CVM lists in one application. For example, one is for domestic transactions and the other is for international transactions.				
Issuer action code (IAC) - decline	b	'9F0E'	5	D8 50 80 00 00 Refer to Section 5.1.2.5 for details	Specifies the conditions for declining the transaction directly without going online.				
Issuer action code (IAC) - online	b	'9F0F'	5	04 28 3C F8 00 Refer to Section 5.1.2.5 for details	Specifies the conditions for transaction online submission.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Issuer action code (IAC) - default	b	'9F0D'	5	04 20 3C A8 00 Refer to Section 5.1.2.5 for details	Specifies Conditions for transaction declining when the transaction request online but terminal cannot complete online escalation.				
Log entry of electronic cash credit for load	b	'DF4D'	2	0C 0A	Provides SFI of log files and number of log file records, <i>UICS Specification</i> has provided the recommendation values: 0C 0A Byte 1: SFI of credit for load log file is 12 (decimal) Byte 2: Number of records in credit for load log file is 10 (decimal)				
Log format of electronic cash credit for load	b	'DF4F'	Var.	Refer to Section 5.1.2.6 for details	List data object tag and length in log record				

5.1.2.2 Application Interchange Profile

‘7D00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	DDA supported
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	1	Support CDA
2	8-1	0000 0000	RFU

5.1.2.3 Processing Option Data Object List

‘9F7A 01 9F02 06 5F2A 02’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Electronic Cash Terminal Supporting Indicator	9F7A	1
Authorized Amount	9F02	6
Transaction Currency Code	5F2A	2

5.1.2.4 Cardholder Verification Method List

‘0000 0000 0000 0000 1F03’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0001 1111 0000 0011	CVM not required	1	If supported by the terminal	Fail Cardholder verification

5.1.2.5 Issuer Action Code (Decline, Online and Default)

‘D8 50 80 00 00’ hexadecimal (IAC-decline)

‘04 28 3C F8 00’ hexadecimal (IAC-online)

‘04 20 3C A8 00’ hexadecimal (IAC-default)

Condition	Byte Bit 8-Bit 1	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
ODA not performed	Byte 1	1	0	0
SDA failed		1	0	0
IC card data missing		0	0	0
Card appeared in terminal exception file		1	0	0
DDA failed		1	0	0
CDA failed		0	1	1
RFU		00	00	00
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		1	0	0

Condition	Byte Bit 8-Bit 1	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
Application not effective yet	Byte 2	0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
CVM fails	Byte 3	1	0	0
Unknown CVM		0	0	0
Times of PIN try exceeds the limit		0	1	1
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad exists, but no PIN is entered		0	1	1
Online PIN entered		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0

Condition	Byte Bit 8-Bit 1	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication failed		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0
RFU		0000	0000	0000

5.1.2.6 Log Format of E-cash Load

‘9A 03 9F21 03 9F1A 02 9F4E 14 9F36 02’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
------------------	-----	--------

Transaction date	9A	3
Transaction time	9F21	3
Terminal country code	9F1A	2
Merchant name and location	9F4E	20
Application Transaction Counter	9F36	2

5.1.3 Low-value Payment Based on qUICS (Contactless) – fDDA 01, Cryptogram Version Number 17

5.1.3.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Card Additional Processes	b	'9F68'	4	'81 00 D0 20' if CVM is online PIN; '81 00 10 20' if CVM is signature. Refer to Section 5.1.3.4 for details.	Indicates card processing requirements and parameter selection				
Card Internal Indicator	b	-	2	The initial setting is 0	To control the card internal processing for qUICS cards.				
Card Transaction Qualifiers	b	'9F6C'	2	The initial setting is '00 00' Please refer to Section 5.1.3.5	Indicates CVM required by the card to the terminal.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Available Offline Purchase Amount	n	'9F5D'	6	The initial setting is 1	A compute field, for the terminal to displaying the amount available for offline transactions, or for issuer risk management.				
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 5.1.3.2 for details	Indicates the functions supported by the card in this application.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Processing Options Data Objects (PDOL)	b	'9F38'	12	9F66 04 9F02 06 9F37 04 5F2A 02 Please refer to Section 5.1.3.3 for details.	The data sent by the terminal to the card, included in GET PROCESSING OPTIONS command, containing terminal data objects (tags and lengths)				
Issuer application data	b	'9F10'	Var.	Refer to Section 5.1.3.6 for details	Specific application data that need to be sent to the Issuer in an online transaction.				
Card Authorization Related Data		'9F69'	8~16	Initialized as 01 00 00 00 00 00 00 00	Input data for dynamic signature of fDDA01.				

5.1.3.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	Support fDDA
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	00000 000	RFU

5.1.3.3 Processing Option Data Object List

‘9F66 04 9F02 06 9F37 04 5F2A 02’ hexadecimal

Data Object Name	Tag	Length
Terminal Transaction Qualifiers	9F66	4
Authorized Amount	9F02	6
Unpredictable Number	9F37	4
Transaction Currency Code	5F2A	2

5.1.3.4 Card Additional Processes

‘81 00 D0 20’ hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	1	Low-value payment checking supported
1	7	0	Neither low-value payment nor CTTA checking supported
1	6	0	RFU
1	5	0	New card checking not supported
1	4	0	PIN Try Limit Exceeded checking not supported
1	3	0	Non-currency matching offline transaction not supported
1	2	0	The card does not prefer to choose contact IC card online transaction first
1	1	1	Return offline purchase amount available.
2	8	0	RFU
2	7	0	Non-currency-matching transaction allowed
2	6	0	If it is a new card and card reader only supports offline transactions, not decline the transaction.
2	5-1	00000	RFU
3	8	1	Currency-matching transactions support online PIN
3	7	1	Non-currency-matching transactions support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU

Byte	Bit	Value	Definition
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

‘81 00 10 20’ hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	1	Low-value payment checking supported
1	7	0	Neither Low-value payment nor CTTA checking supported
1	6	0	RFU
1	5	0	New card checking not supported
1	4	0	PIN Try Limit Exceeded checking not supported
1	3	0	Non-currency matching offline transaction not supported
1	2	0	The card does not prefer to choose contact IC card online transaction first
1	1	1	Return offline purchase amount available.
2	8	0	RFU
2	7	0	Non-currency-matching transaction allowed
2	6	0	If it is a new card and card reader only supports offline transactions, not decline the transaction.
2	5-1	00000	RFU
3	8	0	Currency-matching transactions do not supporting online PIN

Byte	Bit	Value	Definition
3	7	0	Non-currency-matching transactions do not support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.1.3.5 Card Transaction Qualifiers

‘00 00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	0	If offline data authentication fails and the terminal is capable of online processing, not for online processing
1	5	0	If offline data authentication fails and the terminal supports contact IC cards, not terminate the transaction.
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.1.3.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	17	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm indicator
9	8-1	0A	If issuer self-defined data (IDD) exist, self-defined data length (15 max)
10	8-1	01	IDD ID
11-15	8-1	Initialized as 0	Field indicating amount
16-19	8-1	Initialized as 0	MAC

5.2 Template 12 — Common Electronic Cash (1)

5.2.1 Low-value Payment Data Based on Debit/Credit Application (Contact)- CDA, Signature, Credit E-Cash Load Log

5.2.1.1 Card Data Object

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Application Interchange Profile (AIP)	b	'82'	2	7D00 Refer to Section 5.2.1.2 for details	Indicate the functions supported by Low-value payment for Debit/Credit				
Electronic Cash Balance	n	'9F79'	6	Initialized as 00 00 00 00 00 00	Keeps the remaining amount for offline purchase transactions.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Electronic Cash Balance Upper Limit	n	'9F77'	6	Issuers fill in according to actual situation	Represents the max cumulative amount the cardholder can spend on offline purchases in the electronic cash application, and it is the upper limit for topping up the card.				
Electronic Cash Issuer Authorization Code	an	'9F74'	6	ECC001	The Code on the card, indicating the approval of electronic cash transactions.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Transaction Amount Limit	n	'9F78'	6	Issuers fill in according to actual situation	Upper limit of single transaction amount specified by the card, to control risk of single electronic cash transaction.				
Electronic Cash Reset Amount	n	'9F6D'	6	Issuers fill in according to actual situation	Lower limit of available balance, to initiate card automatic loading.				
Processing Options Data Objects (PDOL)	b	'9F38'	9	9F7A 01 9F02 06 5F2A 02 Please refer to Section 5.2.1.3 for details.	The data sent by the terminal to the card, included in GET PROCESSING OPTIONS command, containing terminal data objects (tags and lengths)				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Cardholder Verification Method (CVM)List	b	'8E'	12	0000 0000 0000 0000 1E03 1F00 Please refer to Section 5.2.1.4 for details.	Lists all CVMs supported by the card in order of priority. Note: there can be more than one CVM lists in one application. For example, one is for domestic transactions and the other is for international transactions.				
Issuer action code (IAC) - decline	b	'9F0E'	5	00 10 80 00 00 Refer to Section 5.2.1.5 for details	Specifies the conditions for declining the transaction directly without going online.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Issuer action code (IAC) - online	b	'9F0F'	5	DC 68 3C F8 00 Refer to Section 5.2.1.5 for details	Specifies the conditions for transaction online submission.				
Issuer action code (IAC) - default	b	'9F0D'	5	DC 60 3C A8 00 Refer to Section 5.2.1.5 for details	Specifies Conditions for transaction declining when the transaction request online but terminal cannot complete online escalation.				
Issuer application data	b	'9F10'	Var.	07__ 01 03 00 00 00 01 0A 01 Refer 5.2.1.6 for details	The application data transported to issuer during an online transaction.				

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Records
Entry of E-Cash Load Log	b	DF4D	2	0C0A	Provide the SFI and number of records in E-Cash load log. UICS recommends the value of 0C0A. Byte 1: SFI of load log, the value is 12 (decimal). Byte 2: Number of records in load log, the value is 10 (decimal).				
Format of E-Cash Load Log	b	DF4F	Var	Refer section 5.2.1.6 for details.	List all the tags and lengths of all data objects in Load Log.				

5.2.1.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	Support fDDA
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	1	support CDA
2	8-1	0000 0000	RFU

5.2.1.3 Processing Option Data Object List

‘9F7A 01 9F02 06 5F2A 02’ hexadecimal

Data Object Name	Tag	Length
Electronic Cash Terminal Supporting Indicator	9F7A	1
Authorized Amount	9F02	6
Transaction Currency Code	5F2A	2

5.2.1.4 Cardholder Verification Method List

‘0000 0000 0000 0000 1E03 1F00’ hexadecimal

CVM code – first eight bytes are 0	CVM	Processing Order	Conditions	Action if CVM Fails
0001 1110 0000 0011	Signature	1	If supported by the terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	2	always	Will not fail.

5.2.1.5 Issuer Action Code (Decline, Online and Default)

'00 10 80 00 00' hexadecimal (IAC - decline)

'DC 68 3C F8 00' hexadecimal (IAC - online)

'DC 60 3C A8 00' hexadecimal (IAC - default)

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA failed		0	1	1
IC card data missing		0	0	0
Card appeared in terminal exception file		0	1	1
DDA failed		0	1	1
CDA failed		0	1	1
RFU		00	00	00

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
Versions of IC card and terminal application are inconsistent	Byte 2	0	0	0
Application expired		0	1	1
Application not effective yet		0	1	1
Card does not allow the service requested		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification failed	Byte 3	1	0	0
Unknown CVM		0	0	0
Times of PIN try exceeds the limit		0	1	1
PIN entry required, but PIN pad not present or not working		0	1	1
PIN entry required, PIN pad exists, but no PIN is entered		0	1	1
Online PIN entered		0	1	1

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Lower consecutive offline transaction limit exceeded		0	1	0
Upper consecutive offline transaction limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant forces transaction online		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication failed		0	0	0
Script processing failed before the final GENERATE AC command		0	0	0
Script processing failed after the final GENERATE AC command		0	0	0

Condition	Byte (bit8~bit1)	Result		
		IAC decline – Decline offline	IAC online – Request online	IAC default – Offline decline if cannot go online
RFU		0000	0000	0000

5.2.1.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derived key index
3	8-1	01	Cryptogram version number
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm id
9	8-1	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	8-1	01	IDD ID
11-15	8-1	Initial setting is 0	Amount field
16-19	8-1	Initial setting is 0	MAC

5.2.1.7 Log Format of E-cash Load

‘9A 03 9F21 03 9F1A 02 9F4E 14 9F36 02’ hexadecimal

(Data object tag and length)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Terminal country code	9F1A	2
Merchant name and location	9F4E	20
Application Transaction Counter	9F36	2

5.2.2 Low-value Payment Based on qUICS (Contactless) - fDDA01, Cryptogram Version Number 01, Contactless Transaction Log**5.2.2.1 Card Data Object**

Data Name	Format	Tag	Length (bytes)	Value (Hex)	Description	Template Default Setting	Issuer General Data	Card/ Cardholder Specific Data	Data Stored in File Records
Card Additional Processes	b	'9F68'	4	'91 10 D0 20' if CVM is online PIN; '91 10 10 20' if CVM is signature. Refer to Section 5.2.2.4 for details.	Indicates card processing requirements and parameter selection				

Data Name	Format	Tag	Length (bytes)	Value (Hex)	Description	Template Default Setting	Issuer General Data	Card/ Cardholder Specific Data	Data Stored in File Records
Card CVM Limit	n	'9F6B'	6	Issuers fill in according to actual situation	If it appears, it indicates that currency type of the card matches that of the terminal and a contactless transaction amount exceeds the value, thus CVM provided by the card required. Currently, qUICS supports two CMV: online PIN and Signature.				
Card Internal Indicator	b	-	2	Initialized to 0	To control the card internal processing for qUICS cards.				

Data Name	Format	Tag	Length (bytes)	Value (Hex)	Description	Template Default Setting	Issuer General Data	Card/ Cardholder Specific Data	Data Stored in File Records
Card Transaction Qualifiers	b	'9F6C'	2	Initialized to 00 00 Please refer to Section 5.2.2.5	Indicates CVM required by the card to the terminal.				
Available Offline Purchase Amount	n	'9F5D'	6	Initialized as 1	A compute field, for the terminal to displaying the amount available for offline transactions, or for issuer risk management.				
Application Interchange Profile (AIP)	b	'82'	2	7C00 Refer to Section 5.2.2.2 for details	Indicates the functions supported by the card in this application.				

Data Name	Format	Tag	Length (bytes)	Value (Hex)	Description	Template Default Setting	Issuer General Data	Card/ Cardholder Specific Data	Data Stored in File Records
Processing Options Data Objects (PDOL)	b	'9F38'	24	9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 Please refer to Section 5.2.2.3 for details.	The data sent by the terminal to the card, included in GET PROCESSING OPTIONS command, containing terminal data objects (tags and lengths)				
Issuer Application Data	b	'9F10'	Var.	Please refer to Section 5.2.2.6	The data transported to Issuer in an online transaction.				
Card Authorization Related data		'9F69'	8~16	Initialized as 01 00 00 00 00 00 00 00	Input data for dynamic signature of fDDA01.				

Data Name	Format	Tag	Length (bytes)	Value (Hex)	Description	Template Default Setting	Issuer General Data	Card/ Cardholder Specific Data	Data Stored in File Records
Card Additional Function Indicator	b	'DF61'	Var	40 Please refer to 5.2.2.7	Set Byte 1 bit 7 to 1 means card supports ODA for online authorization				

5.2.2.2 Application Interchange Profile

‘7C00’ hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	SDA supported
1	6	1	Support fDDA
1	5	1	Cardholder verification supported
1	4	1	Terminal risk management supported
1	3	1	Issuer authentication supported
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	00000 000	RFU

5.2.2.3 Processing Option Data Object List

‘9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04’
hexadecimal

Data Object Name	Tag	Length
Terminal Transaction Qualifiers	9F66	4
Authorized Amount	9F02	6
Other Amounts	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result (TVR)	95	5

Data Object Name	Tag	Length
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4

5.2.2.4 Card Additional Processes

‘91 10 D0 20’ hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	1	Low-value payment checking supported
1	7	0	Neither Low-value payment nor CTIA checking supported
1	6	0	RFU
1	5	1	New card checking supported
1	4	0	PIN Try Limit Exceeded checking supported
1	3	0	Non-currency-matching offline transaction not supported
1	2	0	The card does not prefer to choose contact IC card online transaction first
1	1	1	Return offline purchase amount available.
2	8	0	RFU
2	7	0	Non-currency-matching transaction allowed
2	6	0	If it is a new card and card reader only supports offline transactions, not decline the transaction.
2	5	1	Card Transaction Log

Byte	Bit	Value	Definition
2	4-1	00000	RFU
3	8	1	Currency-matching transactions support online PIN
3	7	1	Non-currency-matching transactions support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

‘91 10 10 20’ hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	1	Low-value payment checking supported
1	7	0	Neither Low-value payment nor CTTA checking supported
1	6	0	RFU
1	5	1	New card checking supported
1	4	0	PIN Try Limit Exceeded checking supported
1	3	0	Non-currency matching offline transaction not supported

Byte	Bit	Value	Definition
1	2	0	The card does not prefer to choose contact IC card online transaction first
1	1	1	Return offline purchase amount available.
2	8	0	RFU
2	7	0	Non-currency-matching transaction allowed
2	6	0	If it is a new card and card reader only supports offline transactions, not decline the transaction.
2	5	1	Card Transaction Log
2	4-1	00000	RFU
3	8	0	Currency-matching transactions do not support online PIN
3	7	0	Non-currency-matching transactions do not support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.2.2.5 Card Transaction Qualifiers

'00 00' hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	1	If offline data authentication fails and the terminal is capable of online processing, for online processing
1	5	0	If offline data authentication fails and the terminal supports contact IC cards, not terminate the transaction.
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.2.2.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1	0A	If issuer self-defined data exist, self-defined data length (15 max)
10	8-1	01	IDD ID
11-15	8-1	Initialized as 0	Field indicating amount

Byte	Bit	Hexadecimal Initial Value	Conditions
16-19	8-1	Initialized as 0	MAC

5.2.2.7 Card Additional Function Indicator

Byte	Bit	Hexadecimal Initial Value	Definition
1	8	0	Does not support extended application record data R-MAC protection
	7	1	ODA for online authorization supported
	6	0	RFU
	5	0	Does not support credit-mode low-value payment
	4	0	Does not support dormant card management
	3	0	Does not support bilateral security authentication
	2	0	Does not support fee deduction by segment, does not support offline pre-authorization
	1	0	Does not only support fee deduction by segment

5.3 Template 13 — Common Electronic cash (2)

5.3.1 Low-value Payment Based on Debit/Credit (Contact) - CDA, Signature, Load Log

5.3.1.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Application interchange profile (AIP)	b	'82'	2	7D00 For details see 5.3.1.2	Low-value Payment supported functionality based on debit/credit				
Electronic cash balance	n	'9F79'	6	Initial setting is 00 00 00 00 00 00	Stores remaining amount available for offline purchases.				
Electronic cash balance upper limit	n	'9F77'	6	Issuers fill in according to actual situation	Represents the max cumulative amount the cardholder can spend on offline purchases in the electronic cash application, and it is the upper limit for topping up the card.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Electronic cash issuer authorization code	an	'9F74'	6	ECC001	Code used on card to label approved electronic cash transactions.				
Electronic cash single transaction limit	n	'9F78'	6	Issuers fill in according to actual situation	Upper limit for the amount of any single electronic cash transaction on the card, used to control single electronic cash transaction risk.				
Electronic cash reset threshold	n	'9F6D'	6	Issuers fill in according to actual situation	Usable balance lower limit that triggers card auto top-up.				
Process Options data object list (PDOL)	b	'9F38'	9	9F7A 01 9F02 06 5F2A 02 For details see 5.3.1.3	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Cardholder verification method (CVM) list	b	'8E'	12	0000 0000 0000 0000 1E03 1F00 For details see 5.3.1.4	List out all card application supported CVMs in order of priority.. Note: An application can have multiple CVM tables, for example one for domestic transactions, and one international transaction.				
Issuer action code (IAC) - Reject	b	'9F0E'	5	00 10 80 00 00 For details see 5.3.1.5	Designate conditions where transactions are directly rejected without going online.				
Issuer action code (IAC) - Online	b	'9F0F'	5	DC 68 3C F8 00 For details see 5.3.1.5	Designate condition for transaction to be submitted online.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Issuer action code (IAC) - Default	b	'9F0D'	5	DC 60 3C A8 00 For details see 5.3.1.5	Designate conditions for transaction to be rejected where online request is made but online submission could not be completed.				
Issuer application data	b	'9F10'	Var.	For details see 5.3.1.6	In an online transaction, all custom application data that must be transferred to the issuer.				
Electronic cash circular log entry	b	DF4D	2	0C 0A	Provide circular log file SFI and file record count, UICS specification provides recommended value: 0C 0A Byte 1: Load log file SFI, 12 (decimal) Byte 2: Load log file record count, 10 (decimal)				
Electronic cash circular log format	b	DF4F	Var	For details see 5.3.1.6	Lists tags and lengths of data objects in log records.				

5.3.1.2 Application Interchange Profile

'7D00' hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports DDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	1	Supports CDA
2	8-1	0000 0000	RFU

5.3.1.3 Process Options Data Object List

'9F7A 01 9F02 06 5F2A 02' hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Electronic cash terminal support indicator	9F7A	1
Authorized amount	9F02	6
Transaction currency code	5F2A	2

5.3.1.4 Cardholder Verification Method List

'0000 0000 0000 0000 1E03 1F00' Hexadecimal

CVM code - first 8 bytes are 0	CVM	Processing Order	Condition	Action if CVM fails
0001 1110 0000 0011	Signature	1	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	2	Always	Would never fail

5.3.1.5 Issuer Action Code (Reject, Online, and Default)

'00 10 80 00 00' Hexadecimal (IAC - Reject)

DC 68 3C F8 00' Hexadecimal (IAC - Online)

DC 60 3C A8 00' Hexadecimal (IAC - Default)

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline decline	IAC Online - Request Online	IAC Default - Offline decline if cannot go online
ODA not performed	Byte 1	0	1	1
SDA failed		0	1	1
IC card data incomplete		0	0	0
Card appears in terminal abnormal file		0	1	1
DDA failed		0	1	1
CD) failed		0	1	1
RFU		00	00	00

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline decline	IAC Online - Request Online	IAC Default - Offline decline if cannot go online
IC card and terminal application version mismatch	Byte 2	0	0	0
Application has expired		0	1	1
Application is not yet effective		0	1	1
Card does not permit requested service		1	0	0
New card		0	1	0
RFU		000	000	000
CVM failed	Byte 3	1	0	0
Unknown CVM		0	0	0
PIN retry limit exceeded		0	1	1
Request PIN entry but encryption keyboard does not exist or is non-functional		0	1	1
Request PIN entry, encryption keyboard exists, but PIN not entered		0	1	1

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline decline	IAC Online - Request Online	IAC Default - Offline decline if cannot go online
Enter online PIN		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Consecutive offline transactions lower limit exceeded		0	1	0
Consecutive offline transactions upper limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant requests online processing		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication failure		0	0	0
Script processing failure prior to last GENERATE AC command		0	0	0

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline decline	IAC Online - Request Online	IAC Default - Offline decline if cannot go online
Script processing failure after last GENERATE AC command		0	0	0
RFU		0000	0000	0000

5.3.1.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derived key index
3	8-1	01	Cryptogram version number
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm id
9	8-1	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	8-1	01	IDD ID
11-15	8-1	Initial setting is 0	Amount field
16-19	8-1	Initial setting is 0	MAC

5.3.1.7 Load Log Format

'9A 03 9F21 03 9F1A 02 9F4E 14 9F36 02 Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Terminal country code	9F1A	2
Merchant name and location	9F4E	20
Application Transaction Counter (ATC)	9F36	2

5.3.2 Low-value Payment Based on qUICS (Contactless) - fDDA01, Cryptogram Version 17

5.3.2.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Additional Processes	b	'9F68'	4	'91 00 D0 20' if CVM is online PIN; '91 00 10 20' if CVM is signature. Refer to Section 5.3.2.4 for details.	Points to card processing requirements and parameter selection				
Card CVM Limit	n	'9F6B'	6	Issuers fill in according to actual situation	If it appears this indicates that when card and terminal currency types match and a contactless transaction exceeds this value, then card must provide CVM. Currently qUICS supports two types of CVM: online PIN and signature.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Internal Indicator	b	–	2	Initial setting is 0	Used to control qUICS card internal process.				
Card Transaction Qualifiers	b	‘9F6C’	2	Initial setting is 00 00 For details see 5.3.2.5	Primarily used to indicate card required CVM to the terminal.				
Available Offline Purchase Amount	n	‘9F5D’	6	Initial setting is 1	A calculation field, can be used by terminal to display available offline amount on card, or used for issuer risk control.				
Application Interchange Profile (AIP)	b	‘82’	2	7C00 For details see 5.3.2.2	Describes the application functionality supported by the card				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Process Options Data Object List (PDOL)	b	'9F38'	12	9F66 04 9F02 06 9F37 04 5F2A 02 For details see 5.3.2.3	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				
Issuer Application Data	b	'9F10'	Var.	For details see 5.3.2.6	In an online transaction, all custom application data that must be transferred to the issuer.				
Card authentication related data		'9F69'	8~16	Initial setting is 01 00 00 00 00 00 00 00	'01' version fDDA dynamic signature input data				
Card Additional Function Indicator	b	'DF61'	Var	40 Please refer to 5.3.2.7	Set Byte 1 bit 7 to 1 means card supports ODA for online authorization				

5.3.2.2 Application Interchange Profile

‘7C00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports fDDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000 0000	RFU

5.3.2.3 Process Options Data Object List

‘9F66 04 9F02 06 9F37 04 5F2A 02’ Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Terminal transaction qualifiers	9F66	4
Authorized amount	9F02	6
Unpredictable number	9F37	4
Transaction currency code	5F2A	2

5.3.2.4 Card Additional Processes

‘91 00 D0 20’ Hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Supports PIN retry limit exceeded check
1	3	0	Does not permit non-currency-matching offline transactions
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-matching-currency transactions allowed
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	0	Fast debit/credit transaction does not record in transaction log
2	4-1	0000	RFU
3	8	1	Currency-matching transactions support online PIN
3	7	1	Non-currency-matching transactions support online PIN
3	6	0	For non-currency-matching transactions, CVM is not required by the card
3	5	1	Signature supported

Byte	Bit	Value	Definition
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

'91 00 10 20' Hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Supports PIN retry limit exceeded check
1	3	0	Does not permit non-currency-matching offline transactions
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-matching-currency transactions allowed
2	6	0	If new card and card reader only supports offline then do not reject transaction

Byte	Bit	Value	Definition
2	5	0	Fast debit/credit transactions do not record in transaction log
2	4-1	0000	RFU
3	8	0	Currency-matching transactions do not support online PIN
3	7	0	Non-currency-matching transactions do not support online PIN
3	6	0	For non-currency-matching transactions, CVM is not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.3.2.5 Card Transaction Qualifiers

‘00 00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	0	If offline data authentication fails and terminal can connect online, online connection not required
1	5	0	If offline data authentication fails and terminal supports contact UICS, do not terminate transaction

Byte	Bit	Value	Definition
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.3.2.6 Issuer Application Data

Byte	Hexadecimal Initial Value	Condition
1	07	Length
2	Issuers fill in according to actual situation	Derived key index
3	17	Cryptogram version number
4-7	03 00 00 00	Card verification result (CVR)
8	01	Algorithm id
9	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	01	IDD ID
11-15	Initial setting is 0	Amount field
16-19	Initial setting is 0	MAC

5.3.2.7 Card Additional Function Indicator

Byte	Bit	Hexadecimal Initial Value	Definition
1	8	0	Does not support extended application record data R-MAC protection
	7	1	ODA for online authorization supported
	6	0	RFU
	5	0	Does not support credit-mode low-value payment

Byte	Bit	Hexadecimal Initial Value	Definition
	4	0	Does not support dormant card management
	3	0	Does not support bilateral security authentication
	2	0	Does not support fee deduction by segment, does not support offline pre-authorization
	1	0	Does not support fee deduction by segment

5.4 Template 14 — Electronic Cash that Support Dual Currency

5.4.1 Low-value Payment Based on Debit/Credit (Contact) - CDA, Signature, Load Log

5.4.1.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Application interchange profile (AIP)	b	'82'	2	7D00 <i>For details see 5.4.1.2</i>	Low-value Payment supported functionality based on debit/credit				
Electronic cash balance	n	'9F79'	6	Initial setting is 00 00 00 00 00 00	Stores remaining amount available for offline purchases.				
Electronic cash balance upper limit	n	'9F77'	6	Issuers fill in according to actual situation	Represents the max cumulative amount the cardholder can spend on offline purchases in the electronic cash application, and it is the upper limit for topping up the card.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Electronic cash issuer authorization code	an	'9F74'	6	ECC001	Code used on card to label approved electronic cash transactions.				
Electronic cash single transaction limit	n	'9F78'	6	Issuers fill in according to actual situation	Upper limit for the amount of any single electronic cash transaction on the card, used to control single electronic cash transaction risk.				
Electronic cash reset threshold	n	'9F6D'	6	Issuers fill in according to actual situation	Usable balance lower limit that triggers card auto top-up.				
Process Options Data Object List (PDOL)	b	'9F38'	9	9F7A 01 9F02 06 5F2A 02 <i>For details see 4.4.1.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Cardholder verification method (CVM) list	b	'8E'	12	0000 0000 0000 0000 1E03 1F00 <i>For details see 5.4.1.4</i>	List out all card application supported CVMs in order of priority. Note: An application can have multiple CVM tables, for example one for domestic transactions, and one international transaction.				
Issuer action code (IAC) - Reject	b	'9F0E'	5	00 10 80 00 00 <i>For details see 5.4.1.5</i>	Designate conditions where transactions are directly rejected without going online				
Issuer action code (IAC) - Online	b	'9F0F'	5	DC 68 3C F8 00 <i>For details see 5.4.1.5</i>	Designate condition for transaction to be submitted online.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Issuer action code (IAC) - Default	b	'9F0D'	5	DC 60 3C A8 00 <i>For details see 5.4.1.5</i>	Designate conditions for transaction to be rejected where online request is made but online submission could not be completed.				
Issuer Application Data	b	'9F10'	Var.	<i>For details see 5.4.1.6</i>	In an online transaction, all custom application data that must be transferred to the issuer.				
Electronic cash circular log entry	b	DF4D	2	0C 0A	Provide circular log file SFI and file record count, UICS specification provides recommended value: 0C 0A Byte 1: Load log file SFI, 12 (decimal) Byte 2: Load log file record count, 10 (decimal)				
Electronic cash circular log format	b	DF4F	Var	<i>For details see 5.4.1.6</i>	Lists tags and lengths of data objects in log records.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Second currency electronic cash application currency code	n4	DF71	2	Issuers fill in according to actual situation					
Secondary currency electronic cash balance	n12	DF79	6	Issuers fill in according to actual situation					
Secondary currency electronic cash balance upper limit	n12	DF77	6	Issuers fill in according to actual situation					
Secondary currency electronic cash single transaction limit	n12	DF78	6	Issuers fill in according to actual situation					

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Secondary currency electronic cash reset threshold	n12	DF76	6	Issuers fill in according to actual situation					

5.4.1.2 Application Interchange Profile

'7D00' Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports DDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	1	Supports CDA
2	8-1	0000 0000	RFU

5.4.1.3 Process Options Data Object List

'9F7A 01 9F02 06 5F2A 02' Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Electronic cash terminal support indicator	9F7A	1
Authorized amount	9F02	6
Transaction currency code	5F2A	2

5.4.1.4 Cardholder Verification Method List

'0000 0000 0000 0000 1E03 1F00' Hexadecimal

CVM code - first 8 bytes are 0	CVM	Processing Order	Condition	Action if CVM fails
0001 1110 0000 0011	Signature	1	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	2	Always	Will not fail

5.4.1.5 Issuer Action Code (Reject, Online, and Default)

'00 10 80 00 00' Hexadecimal (IAC - Reject)

'DC 68 3C F8 00' Hexadecimal (IAC - Online)

'DC 60 3C A8 00' Hexadecimal (IAC - Default)

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
No offline data authentication	Byte 1	0	1	1
Offline SDA failure		0	1	1
IC card data incomplete		0	0	0
Card appears in terminal abnormal file		0	1	1
Offline DDA failure		0	1	1
Combined dynamic data authentication/ application code generation (CDA) failure		0	1	1

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
RFU		00	00	00
IC card and terminal application version mismatch	Byte 2	0	0	0
Application has expired		0	1	1
Application is not yet effective		0	1	1
Card does not permit requested service		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification failure	Byte 3	1	0	0
Unknown CVM		0	0	0
PIN retry limit exceeded		0	1	1
Request PIN entry but encryption keyboard does not exist or is non-functional		0	1	1

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
Request PIN entry, encryption keyboard exists, but PIN not entered		0	1	1
Enter online PIN		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Consecutive offline transactions lower limit exceeded		0	1	0
Consecutive offline transactions upper limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant requests online processing		0	1	1
RFU		000	000	000
Use default TDOL		0	0	0
Issuer authentication failure	Byte 5	0	0	0

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
Script processing failure prior to last GENERATE AC command		0	0	0
Script processing failure after last GENERATE AC command		0	0	0
RFU		0000	0000	0000

5.4.1.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derived key index
3	8-1	01	Cryptogram version number
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm id
9	8-1	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	8-1	01	IDD ID
11-15	8-1	Initial setting is 0	Amount field

Byte	Bit	Hexadecimal Initial Value	Condition
16-19	8-1	Initial setting is 0	MAC

5.4.1.7 Load Log Format

'9A 03 9F21 03 9F1A 02 9F4E 14 9F36 02 Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Terminal country code	9F1A	2
Merchant name and location	9F4E	20
Application Transaction Counter (ATC)	9F36	2

5.4.2 Low-value Payment Based on qUICS (contactless) - fDDA01, Application Cryptogram Version 01

5.4.2.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Additional Processes	b	'9F68',	4	'91 00 D0 20' if CVM is online PIN; '91 00 10 20' if CVM is signature. Refer to Section 5.4.2.4 for details.	Points to card processing requirements and parameter selection				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card CVM Limit	n	'9F6B ,	6	Issuers fill in according to actual situation	If it appears this indicates that when card and terminal currency types match and a contactless transaction exceeds this value, then card must provide CVM. Currently qUICS supports two types of CVM: online PIN and signature.				
Card internal indicator	b	—	2	Initial setting is 0	Used to control qUICS card internal process.				
Card Transaction Qualifiers	b	'9F6C ,	2	Initial setting is 00 00 <i>For details see 5.4.2.5</i>	Primarily used to indicate card required CVM to the terminal.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Available Offline Purchase Amount	n	'9F5D'	6	Initial setting is 1	A calculation field, can be used by terminal to display available offline amount on card, or used for issuer risk control.				
Application interchange profile (AIP)	b	'82'	2	7C00 <i>For details see 5.4.2.2</i>	Describes the application functionality supported by the card				
Process Options Data Object List (PDOL)	b	'9F38',	24	9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 <i>For details see 5.4.2.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Issuer Application Data	b	'9F10',	Var.	For details see 5.4.2.6	In an online transaction, all custom application data that must be transferred to the issuer.				
Second currency card CVM limit	n12	DF72	6	Issuers fill in according to actual situation					
Card authentication related data		'9F69',	8~16	Initial setting is 01 00 00 00 00 00 00 00	'01' version fDDA dynamic signature input data				
Card Additional Function Indicator	b	'DF61'	Var	C0 Please refer to 5.4.2.7	Set Byte 1 bit 7 to 1 means card supports ODA for online authorization				

5.4.2.2 Application Interchange Profile

'7C00' Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports fDDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	00000000	RFU

5.4.2.3 Process Options Data Object List

'9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04'
Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Terminal Transaction Qualifiers	9F66	4
Authorized Amount	9F02	6
Other Amount	9F03	6
Terminal Country Code	9F1A	2

Data Object Name	Tag	Length
Terminal Verification Result (TVR)	95	5
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4

5.4.2.4 Card Additional Processes

‘91 00 D0 20’ Hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Does not support PIN retry limit exceeded check
1	3	0	Does not permit non-currency-matching offline transactions
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-currency-matching currency transactions allowed

Byte	Bit	Value	Definition
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	0	Fast debit/credit transaction does not record in transaction log
2	4-1	0000	RFU
3	8	1	Currency-matching transactions support online PIN
3	7	1	Non-currency-matching transactions support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

‘91 00 10 20’ Hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Does not support PIN retry limit exceeded check

Byte	Bit	Value	Definition
1	3	0	Does not permit non-currency-matching offline transactions
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-currency-matching currency transactions allowed
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	0	Fast debit/credit transaction does not record in transaction log
2	4-1	0000	RFU
3	8	0	Currency-matching transactions do not support online PIN
3	7	0	Non-currency-matching transactions do not support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.4.2.5 Card Transaction Qualifiers

‘00 00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	0	If offline data authentication fails and terminal can connect online, online connection not required
1	5	0	If offline data authentication fails and terminal supports contact UICS, do not terminate transaction
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.4.2.6 Issuer Application Data

Byte	Hexadecimal Initial Value	Condition
1	07	Length
2	Issuers fill in according to actual situation	Derived key index
3	01	Cryptogram version number
4-7	03 00 00 00	Card verification result (CVR)
8	01	Algorithm id
9	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)

Byte	Hexadecimal Initial Value	Condition
10	01	IDD ID
11-15	Initial setting is 0	Amount field
16-19	Initial setting is 0	MAC

5.4.2.7 Card Additional Function Indicator

Byte	Bit	Hexadecimal Initial Value	Definition
1	8	1	Support extended application record data R-MAC protection
	7	1	ODA for online authorization supported
	6	0	RFU
	5	0	Does not support credit-mode low-value payment
	4	0	Does not support dormant card management
	3	0	Does not support bilateral security authentication
	2	0	Does not support fee deduction by segment, does not support offline pre-authorization
	1	0	Does not support fee deduction by segment

5.5 Template 15 — Electronic Cash that Support Extended Applications

5.5.1 Low-value Payment Based on Debit/Credit (Contact) - CDA, Signature, Load Log

5.5.1.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Application interchange profile (AIP)	b	'82'	2	7D00 <i>For details see 5.5.1.2</i>	Low-value Payment supported functionality based on debit/credit				
Electronic cash balance	n	'9F79'	6	Initial setting is 00 00 00 00 00	Stores remaining amount available for offline purchases.				
Electronic cash balance upper limit	n	'9F77'	6	Issuers fill in according to actual situation	Represents the max cumulative amount the cardholder can spend on offline purchases in the electronic cash application, and it is the upper limit for topping up the card.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Electronic cash issuer authorization code	an	'9F74'	6	ECC001	Code used on card to label approved electronic cash transactions.				
Electronic cash single transaction limit	n	'9F78'	6	Issuers fill in according to actual situation	Upper limit for the amount of any single electronic cash transaction on the card, used to control single electronic cash transaction risk.				
Electronic cash reset threshold	n	'9F6D'	6	Issuers fill in according to actual situation	Usable balance lower limit that triggers card auto top-up.				
Process Options Data Object List (PDOL)	b	'9F38'	9	9F7A 01 9F02 06 5F2A 02 <i>For details see 5.5.1.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Cardholder verification method (CVM) list	b	'8E',	12	0000 0000 0000 0000 1E03 1F00 <i>For details see 5.5.1.4</i>	List out all card application supported CVMs in order of priority.. Note: An application can have multiple CVM tables, for example one for domestic transactions, and one international transaction.				
Issuer action code (IAC) - Reject	b	'9F0E'	5	00 10 80 00 00 <i>For details see 5.5.1.5</i>	Designate conditions where transactions are directly rejected without going online.				
Issuer action code (IAC) - Online	b	'9F0F'	5	DC 68 3C F8 00 <i>For details see 5.5.1.5</i>	Designate condition for transaction to be submitted online.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Issuer action code (IAC) - Default	b	'9F 0D'	5	DC 60 3C A8 00 <i>For details see 5.5.1.5</i>	Designate conditions for transaction to be rejected where online request is made but online submission could not be completed.				
Issuer Application Data	b	'9F 10'	Var.	<i>For details see 5.2.1.6</i>	In an online transaction, all custom application data that must be transferred to the issuer.				
Electronic cash circular log entry	b	DF 4D	2	0C 0A	Provide circular log file SFI and file record count, UICS specification provides recommended value: 0C 0A Byte 1: Load log file SFI, 12 (decimal) Byte 2: Load log file record count, 10 (decimal)				
Electronic cash circular log format	b	DF 4F	Var	<i>For details see 5.5.1.6</i>	Lists tags and lengths of data objects in log records.				

5.5.1.2 Application Interchange Profile

'7D00' Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports DDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	1	Supports CDA
2	8-1	0000 0000	RFU

5.5.1.3 Process Options Data Object List

'9F7A 01 9F02 06 5F2A 02' Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Electronic cash terminal support indicator	9F7A	1
Authorized amount	9F02	6
Transaction currency code	5F2A	2

5.5.1.4 Cardholder Verification Method List

'0000 0000 0000 0000 1E03 1F00' Hexadecimal

CVM code - first 8 bytes are 0	CVM	Processing Order	Condition	If this CVM fail
0001 1110 0000 0011	Signature	1	If supported by terminal	Fail Cardholder verification
0001 1111 0000 0000	CVM not required	2	Always	Will not fail

5.5.1.5 Issuer Action Code (Reject, Online, and Default)

'00 10 80 00 00' Hexadecimal (IAC - Reject)

'DC 68 3C F8 00' Hexadecimal (IAC - Online)

'DC 60 3C A8 00' Hexadecimal (IAC - Default)

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
No offline data authentication	Byte 1	0	1	1
Offline SDA failure		0	1	1
IC card data incomplete		0	0	0
Card appears in terminal abnormal file		0	1	1
Offline DDA failure		0	1	1
Combined dynamic data authentication/ application code generation (CDA) failure		0	1	1
RFU		00	00	00
IC card and terminal application version mismatch	Byte 2	0	0	0

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
Application has expired		0	1	1
Application is not yet effective		0	1	1
Card does not permit requested service		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification failure	Byte 3	1	0	0
Unknown CVM		0	0	0
PIN retry limit exceeded		0	1	1
Request PIN entry but encryption keyboard does not exist or is non-functional		0	1	1
Request PIN entry, encryption keyboard exists, but PIN not entered		0	1	1
Enter online PIN		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Consecutive offline transactions lower limit exceeded		0	1	0

Condition	Byte (bit8~bit1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
Consecutive offline transactions upper limit exceeded		0	1	1
Transaction is randomly selected for online processing		0	1	0
Merchant requests online processing		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication failure		0	0	0
Script processing failure prior to last GENERATE AC command		0	0	0
Script processing failure after last GENERATE AC command		0	0	0
RFU		0000	0000	0000

5.5.1.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8- 1	07	Length
2	8- 1	Issuers fill in according to actual situation	Derived key index

Byte	Bit	Hexadecimal Initial Value	Condition
3	8-1	01	Cryptogram version number
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm id
9	8-1	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	8-1	01	IDD ID
11-15	8-1	Initial setting is 0	Amount field
16-19	8-1	Initial setting is 0	MAC

5.5.1.7 Load Log Format

‘9A 03 9F21 03 9F1A 02 9F4E 14 9F36 02’ Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Terminal country code	9F1A	2
Merchant name and location	9F4E	20
Application Transaction Counter (ATC)	9F36	2

5.5.2 Low-value Payment Based on qUICS (Contactless) - fDDA01, Cryptogram Version 01, Supports R-MAC Protection

5.5.2.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Additional Processes	b	'9F68'	4	'91 00 D0 20' if CVM is online PIN; '91 00 10 20' if CVM is signature. Refer to Section 5.5.2.4 for details.	Points to card processing requirements and parameter selection				
Card CVM Limit	n	'9F6B'	6	Issuers fill in according to actual situation	If it appears this indicates that when card and terminal currency types match and a contactless transaction exceeds this value, then card must provide CVM. Currently qUICS supports two types of CVM: online PIN and signature.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card internal indicator	b	–	2	Initial setting is 0	Used to control qUICS card internal process.				
Card Transaction Qualifiers	b	‘9F6C’	2	Initial setting is 00 00 <i>For details see 5.5.2.5</i>	Primarily used to indicate card required CVM to the terminal.				
Available Offline Purchase Amount	n	‘9F5D’	6	Initial setting is 1	A calculation field, can be used by terminal to display available offline amount on card, or used for issuer risk control.				
Application interchange profile (AIP)	b	‘82’	2	7C00 <i>For details see 5.5.2.2</i>	Describes the application functionality supported by the card				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Process Options Data Object List (PDOL)	b	'9F38'	27	9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 DF60 01 <i>For details see 5.5.2.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				
Issuer Application Data	b	'9F10'	Var.	<i>For details see 5.5.2.6</i>	In an online transaction, all custom application data that must be transferred to the issuer.				
Card Authentication Related Data		'9F69'	8~16	Initial setting is 01 00 00 00 00 00 00 XX	'01' version fDDA dynamic signature input data Note: Its Byte 8 for this template shall be set as the value of deduction by segment application id (tag 'DF61') Byte 1.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Application access subkey	b	—	16	Issuers fill in according to actual situation	Derived from issuer's industry application access key, each industry (extended application proprietary file) corresponds to one application access key				
Card Additional Function Indicator	v	'DF61'	Var	C2 <i>For details see 5.5.2.7</i>	Card supports fee deduction by segment and offline pre-authorization, also supports extended application data R-MAC protection (currently only defines first byte)				
Fee deduction by segment limit	n	DF62	6	Issuers fill in according to actual situation	Max deposit deduction amount supported for card fee deduction by segment				
Fee deducted by segment	n	DF63	6	Initial setting is 0	Amount already deducted with fee deduction by segment application				

5.5.2.2 Application Interchange Profile

‘7C00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports fDDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	00000000	RFU

5.5.2.3 Process Options Data Object List

‘9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 DF60 01’ Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Terminal Transaction Qualifiers	9F66	4
Authorized Amount	9F02	6
Other Amount	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result (TVR)	95	5

Data Object Name	Tag	Length
Transaction Currency Code	5F2A	2
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4
CAPP Transaction Indicator	DF60	1

5.5.2.4 Card Additional Processes

‘91 00 D0 20’ Hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Does not support PIN retry limit exceeded check
1	3	0	Non-currency-matching offline transactions not allowed
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Permit non-currency-matching transactions

Byte	Bit	Value	Definition
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	0	Fast debit/credit transaction does not record in transaction log
2	4-1	0000	RFU
3	8	1	Currency-matching transactions support online PIN
3	7	1	Non-currency-matching transactions support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

‘91 00 10 20’ Hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check

Byte	Bit	Value	Definition
1	4	0	Does not support PIN retry limit exceeded check
1	3	0	Non-currency-matching offline transactions not allowed
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-currency-matching transactions supported
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	0	Fast debit/credit transaction does not record in transaction log
2	4-1	0000	RFU
3	8	0	Currency-matching transactions do not support online PIN
3	7	0	Non-currency-matching transactions do not support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.5.2.5 Card Transaction Qualifiers

‘00 00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	0	If offline data authentication fails and terminal can connect online, online connection not required
1	5	0	If offline data authentication fails and terminal supports contact UICS, do not terminate transaction
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.5.2.6 Issuer Application Data

Byte	Hexadecimal Initial Value	Condition
1	07	Length
2	Issuers fill in according to actual situation	Derived key index
3	01	Cryptogram version number
4-7	03 00 00 00	Card verification result (CVR)
8	01	Algorithm id
9	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)

Byte	Hexadecimal Initial Value	Condition
10	01	IDD ID
11-15	Initial setting is 0	Amount field
16-19	Initial setting is 0	MAC

5.5.2.7 Card Additional Function Indicator

Byte	Bit	Hexadecimal Initial Value	Description
1	8	1	Support extended application record data R-MAC protection
	7	1	ODA for online authorization supported
	6	0	RFU
	5	0	Credit-mode low-value payment not supported
	4	0	Does not support dormant card management
	3	0	Does not support bilateral security authentication
	2	1	Support both fee deduction by segment functionality and offline pre-authorization functionality
	1	0	Card does not only support fee deduction by segment

5.6 Template 16 — Electronic Cash that Support Enhanced Functionality

5.6.1 Low-value Payment Based on Debit/Credit (Contact) - CDA, Signature, Load Log

5.6.1.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Application Interchange Profile (AIP)	b	'82'	2	7D00 <i>For details see 5.6.1.2</i>	Low-value Payment supported functionality based on debit/credit				
Electronic Cash Balance	n	'9F79'	6	Initial setting is 00 00 00 00 00 00	Stores remaining amount available for offline purchases.				
Electronic Cash Balance Upper Limit	n	'9F77'	6	Issuers fill in according to actual situation	Represents the max cumulative amount the cardholder can spend on offline purchases in the electronic cash application, and it is the upper limit for topping up the card.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Electronic Cash Issuer Authorization Code	an	'9F74'	6	ECC001	Code used on card to label approved electronic cash transactions.				
Electronic Cash Single Transaction Limit	n	'9F78'	6	Issuers fill in according to actual situation	Upper limit for the amount of any single electronic cash transaction on the card, used to control single electronic cash transaction risk.				
Electronic Cash Reset Threshold	n	'9F6D'	6	Issuers fill in according to actual situation	Usable balance lower limit that triggers card auto top-up.				
Process Options Data Object List (PDOL)	b	'9F38'	6	9F02 06 5F2A 02 <i>For details see 5.6.1.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length). (Enhanced micropayment does not support contact electronic cash)				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Cardholder Verification Method (CVM) list	b	'8E'	12	0000 0000 0000 0000 1E03 1F00 <i>For details see 5.6.1.4</i>	List out all card application supported CVMs in order of priority.. Note: An application can have multiple CVM tables, for example one for domestic transactions, and one international transaction.				
Issuer Action Code (IAC) - Reject	b	'9F0E'	5	00 10 80 00 00 <i>For details see 5.6.1.5</i>	Designate conditions where transactions are directly rejected without going online.				
Issuer Action Code (IAC) - Online	b	'9F0F'	5	DC 68 3C F8 00 <i>For details see 5.6.1.5</i>	Designate condition for transaction to be submitted online.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Issuer action code (IAC) - Default	b	'9F0D'	5	DC 60 3C A8 00 <i>For details see 5.6.1.5</i>	Designate conditions for transaction to be rejected where online request is made but online submission could not be completed.				
Issuer Application Data	b	'9F10'	Var.	<i>For details see 5.6.1.6</i>	In an online transaction, all custom application data that must be transferred to the issuer.				
Electronic cash circular log entry	b	DF4D	2	0C 0A	Provide circular log file SFI and file record count, UICS specification provides recommended value: 0C 0A Byte 1: Load log file SFI, 12 (decimal) Byte 2: Load log file record count, 10 (decimal)				
Electronic cash circular log format	b	DF4F	Var	<i>For details see 5.6.1.6</i>	Lists tags and lengths of data objects in log records				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Second currency electronic cash application currency code	n4	DF71	2	Issuers fill in according to actual situation					
Secondary currency electronic cash balance	n12	DF79	6	Issuers fill in according to actual situation					
Secondary currency electronic cash balance upper limit	n12	DF77	6	Issuers fill in according to actual situation					
Secondary currency electronic cash single transaction limit	n12	DF78	6	Issuers fill in according to actual situation					

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Secondary currency electronic cash reset threshold	n12	DF76	6	Issuers fill in according to actual situation					

5.6.1.2 Application Interchange Profile

'7D00' Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports DDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	1	Supports CDA
2	8-1	0000 0000	RFU

5.6.1.3 Process Options Data Object List

'9F02 06 5F2A 02' Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Authorized amount	9F02	6
Transaction currency code	5F2A	2

5.6.1.4 Cardholder Verification Method List

'0000 0000 0000 0000 1E03 1F00' Hexadecimal

CVM code - first 8 bytes are 0	CVM	Processing Order	Condition	If this CVM fail
0001 1110 0000 0011	Signature	1	If supported by terminal	Fail Cardholder Verification
0001 1111 0000 0000	CVM not required	2	Always	Will not fail

5.6.1.5 Issuer Action Code (Reject, Online, and Default)

'00 10 80 00 00' Hexadecimal (IAC - Reject)

'DC 68 3C F8 00' Hexadecimal (IAC - Online)

'DC 60 3C A8 00' Hexadecimal (IAC - Default)

Condition	Byte (bit8~bit 1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
No offline data authentication	Byte 1	0	1	1
Offline SDA failure		0	1	1
IC card data incomplete		0	0	0
Card appears in terminal abnormal file		0	1	1
Offline DDA failure		0	1	1
Combined dynamic data authentication/ application code generation (CDA) failure		0	1	1
RFU		00	00	00
IC card and terminal application version mismatch	Byte 2	0	0	0

Condition	Byte (bit8~bit 1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
Application has expired		0	1	1
Application is not yet effective		0	1	1
Card does not permit requested service		1	0	0
New card		0	1	0
RFU		000	000	000
Cardholder verification failure	Byte 3	1	0	0
Unknown CVM		0	0	0
PIN retry limit exceeded		0	1	1
Request PIN entry but encryption keyboard does not exist or is non-functional		0	1	1
Request PIN entry, encryption keyboard exists, but PIN not entered		0	1	1
Enter online PIN		0	1	1
RFU		00	00	00
Transaction exceeds lower limit	Byte 4	0	1	1
Consecutive offline transactions lower limit exceeded		0	1	0
Consecutive offline transactions upper limit exceeded		0	1	1

Condition	Byte (bit8~bit 1)	Result		
		IAC Reject - Offline Reject	IAC Online - Request Online	IAC Default - If can't go online then reject offline
Transaction is randomly selected for online processing		0	1	0
Merchant requests online processing		0	1	1
RFU		000	000	000
Use default TDOL	Byte 5	0	0	0
Issuer authentication failure		0	0	0
Script processing failure prior to last GENERATE AC command		0	0	0
Script processing failure after last GENERATE AC command		0	0	0
RFU		0000	0000	0000

5.6.1.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Condition
1	8- 1	07	Length
2	8- 1	Issuers fill in according to actual situation	Derived key index
3	8- 1	01	Cryptogram version number

Byte	Bit	Hexadecimal Initial Value	Condition
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm id
9	8-1	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	8-1	01	IDD ID
11-15	8-1	Initial setting is 0	Amount field
16-19	8-1	Initial setting is 0	MAC

5.6.1.7 Load Log Format

'9A 03 9F21 03 9F1A 02 9F4E 14 9F36 02 Hexadecimal

(Data object tags and lengths)

Data Object Name	Tag	Length
Transaction date	9A	3
Transaction time	9F21	3
Terminal country code	9F1A	2
Merchant name and location	9F4E	20
Application Transaction Counter (ATC)	9F36	2

5.6.2 Low-value Payment Based on qUICS (Contactless) - fDDA01, Application Cryptogram Version 01

5.6.2.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Additional Processes	b	'9F68',	4	'91 10 D0 A0' if CVM is online PIN; '91 10 10 A0' if CVM is signature. Refer to Section 5.6.2.4 for details.	Points to card processing requirements and parameter selection				
Card CVM Limit	n	'9F6B',	6	Issuers fill in according to actual situation	If it appears this indicates that when card and terminal currency types match and a contactless transaction exceeds this value, then card must provide CVM. Currently qUICS supports two types of CVM: online PIN and signature.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card internal indicator	b	—	2	Initial setting is 0	Used to control qUICS card internal process.				
Card Transaction Qualifiers	b	'9F6C',	2	Initial setting is 00 00 <i>For details see 5.6.2.5</i>	Primarily used to indicate card required CVM to the terminal.				
Available Offline Purchase Amount	n	'9F5D'	6	Initial setting is 1	A calculation field, can be used by terminal to display available offline amount on card, or used for issuer risk control.				
Application interchange profile (AIP)	b	'82'	2	7C00 <i>For details see 5.6.2.2</i>	Describes the application functionality supported by the card				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Process Options Data Object List (PDOL)	b	'9F38 ,	30	9F6604 9F0206 9F0306 9F1A02 9505 5F2A02 9A03 9C01 9F3704 9F2103 9F4E14 <i>For details see 5.6.2.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				
Issuer Application Data	b	'9F10 ,	Var.	<i>For details see 5.6.2.6</i>	In an online transaction, all custom application data that must be transferred to the issuer.				
Second currency card CVM limit	n12	DF72	6	Issuers fill in according to actual situation					

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Authentication Related Data		'9F69',	8~16	Initial setting is 01 00 00 00 00 00 00 00	'01' version fDDA dynamic signature input data				
Card Additional Function Indicator	b	'DF61'	Var (only defines 1st byte)	4C <i>For details see 5.6.2.7</i>	Card supports dormant card management, supports mutual authentication				

5.6.2.2 Application Interchange Profile

‘7C00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports fDDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	00000000	RFU

5.6.2.3 Process Options Data Object List

‘9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ Hexadecimal (data object tags and lengths)

Data Object Name	Tag	Length
Terminal transaction qualifiers	9F66	4
Authorized amount	9F02	6
Other amount	9F03	6
Terminal country code	9F1A	2
Terminal verification result (TVR)	95	5
Transaction currency code	5F2A	2
Transaction date	9A	3

Data Object Name	Tag	Length
Transaction type	9C	1
Unpredictable number	9F37	4
Transaction time	9F21	3
Merchant name and location	9F4E	14

5.6.2.4 Card Additional Processes

‘91 10 D0 A0’ Hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Does not support PIN retry limit exceeded check
1	3	0	Does not permit mismatching currency offline transactions
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-currency-matching transactions allowed

Byte	Bit	Value	Definition
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	1	Record transaction log for qUICS offline approved transactions
2	4-1	0000	RFU
3	8	1	Currency-matching transactions support online PIN
3	7	1	Non-currency-matching transactions support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU
4	8	1	Mandatory mutual authentication for 1st currency type
4	7	0	Optional mutual authentication for 2nd currency type
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

‘91 10 10 A0’ Hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	1	Supports low-value payment check

Byte	Bit	Value	Definition
1	7	0	Does not support low-value payment and CTTA check
1	6	0	RFU
1	5	1	Supports new card check
1	4	0	Does not support PIN retry limit exceeded check
1	3	0	Does not permit mismatching currency offline transactions
1	2	0	Card does not prefer to choose contact UICS online connection
1	1	1	Return offline purchasing available amount
2	8	0	RFU
2	7	0	Non-currency-matching transactions allowed
2	6	0	If new card and card reader only supports offline then do not reject transaction
2	5	1	Record transaction log for qUICS offline approved transactions
2	4-1	0000	RFU
3	8	0	Currency-matching transactions do not support online PIN
3	7	0	Non-currency-matching transactions do not support online PIN
3	6	0	For non-currency-matching transactions, CVM not required by the card
3	5	1	Signature supported
3	4-1	0000	RFU

Byte	Bit	Value	Definition
4	8	1	Mandatory mutual authentication for 1st currency type
4	7	0	Optional mutual authentication for 2nd currency type
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.6.2.5 Card Transaction Qualifiers

‘00 00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	0	If offline data authentication fails and terminal can connect online, online connection not required
1	5	0	If offline data authentication fails and terminal supports contact UICS, do not terminate transaction
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.6.2.6 Issuer Application Data

Byte	Hexadecimal Initial Value	Condition
1	07	Length

Byte	Hexadecimal Initial Value	Condition
2	Issuers fill in according to actual situation	Derived key index
3	01	Cryptogram version number
4-7	03 00 00 00	Card verification result (CVR)
8	01	Algorithm id
9	0A	Issuer Discretionary Data (IDD) - self-defined data length (max 15)
10	01	IDD ID
11-15	Initial setting is 0	Amount field
16-19	Initial setting is 0	MAC

5.6.2.7 Card Additional Function Indicator

Byte	Bit	Hexadecimal Initial Value	Description
1	8	0	Does not support extended application record data R-MAC protection
	7	1	ODA for online authorization supported
	6	0	RFU
	5	0	Credit-mode low-value payment not supported
	4	1	Card supports dormant card management

Byte	Bit	Hexadecimal Initial Value	Description
	3	1	Card supports bilateral security authentication
	2	0	Does not support fee deduction by segment, does not support offline pre-authorization
	1	0	Does not support fee deduction by segment

5.7 Template 17 — Online Only Template (No E-cash Functions)

5.7.1 Low-value Payment Based on qUICS (Contactless) - fDDA01, Cryptogram Version 01

5.7.1.1 Card Data Object

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card Additional Processes	b	'9F68'	4	'10 20 D0 20' if CVM is online PIN; '10 20 10 20' if CVM is signature. Refer to Section 5.7.1.4 for details.	Points to card processing requirements and parameter selection				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Card CVM Limit	n	'9F6B'	6	Issuers fill in according to actual situation	If it appears this indicates that when card and terminal currency types match and a contactless transaction exceeds this value, then card must provide CVM. Currently qUICS supports two types of CVM: online PIN and signature.				
Card internal indicator	b	—	2	Initial setting is 0	Used to control qUICS card internal process.				
Card Transaction Qualifiers	b	'9F6C'	2	Initial setting is 00 00 <i>For details see 5.7.1.5</i>	Primarily used to indicate card required CVM to the terminal.				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Available Offline Purchase Amount	n	'9F5D'	6	Initial setting is 1	A calculation field, can be used by terminal to display available offline amount on card, or used for issuer risk control.				
Application Interchange Profile (AIP)	b	'82'	2	7C00 <i>For details see 5.7.1.2</i>	Describes the application functionality supported by the card				
Process Options Data Object List (PDOL)	b	'9F38'	24	9F6604 9F0206 9F0306 9F1A02 9505 5F2A02 9A03 9C01 9F3704 <i>For details see 5.7.1.3</i>	Designate data sent to card by terminal via PDOL. Includes terminal data objects (tag and length).				

Data Name	Card Data Format	Tag	Data length (bytes)	Value (hexadecimal)	Description	Template Default Setting	Issuer general data	Card or Cardholder Special Data	Data Stored in File Record
Issuer Application Data	b	'9F10'	Var.	07__ 01 03 00 00 00 01 0A 01 <i>For details see 5.7.1.6</i>	In an online transaction, all custom application data that must be transferred to the issuer.				
Card Authentication Related Data		'9F69'	8~16	Initial setting is 01 00 00 00 00 00 00 00	'01' version fDDA dynamic signature input data				
Card Additional Function Indicator	b	'DF61'	Var	40 Please refer to 5.7.1.7	Set Byte 1 bit 7 to 1 means card supports ODA for online authorization				

5.7.1.2 Application Interchange Profile

‘7C00’ Hexadecimal

Byte	Bit	Value	Definition
1	8	0	RFU
1	7	1	Supports SDA
1	6	1	Supports fDDA
1	5	1	Supports cardholder authentication
1	4	1	Supports terminal risk management
1	3	1	Supports issuer authentication
1	2	0	RFU
1	1	0	CDA not supported
2	8-1	0000000 0	RFU

5.7.1.3 Process Options Data Object List

‘9F66 04 9F02 06 9F03 06 9F1A 02 95 05 5F2A 02 9A 03 9C 01 9F37 04 9F21 03 9F4E 14’ Hexadecimal (data object tags and lengths)

Data Object Name	Tag	Length
Terminal Transaction Qualifiers	9F66	4
Authorized Amount	9F02	6
Other Amount	9F03	6
Terminal Country Code	9F1A	2
Terminal Verification Result (TVR)	95	5
Transaction Currency Code	5F2A	2

Data Object Name	Tag	Length
Transaction Date	9A	3
Transaction Type	9C	1
Unpredictable Number	9F37	4

5.7.1.4 Card Additional Processes

‘10 20 D0 20’ Hexadecimal (CVM is online PIN)

Byte	Bit	Value	Definition
1	8	0	Low-value payment checking not supported
	7	0	Neither Low-value payment nor CTTA checking supported
	6	0	RFU
	5	1	New card checking supported
	4	0	PIN Try Limit Exceeded checking supported
	3	0	Non-currency matching offline transaction not supported
	2	0	The card does not prefer to choose contactIC card online transaction first
	1	0	Do not return offline purchase amount available.
2	8	0	RFU
	7	0	Non-currency-matching transaction not allowed
	6	1	If it is a new card and card reader only supports offline transactions, not decline the transaction.

Byte	Bit	Value	Definition
	5	0	Card Transaction Log
	4-1	00000	RFU
3	8	1	Currency-matching transaction supporting online PIN
	7	1	Non-currency-matching transaction supporting online PIN
	6	0	For non-currency-matching transactions, CVM not required
	5	1	Signature supported
	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

‘10 20 10 20’ Hexadecimal (CVM is signature)

Byte	Bit	Value	Definition
1	8	0	Low-value payment checking not supported
	7	0	Neither Low-value payment nor CTTA checking supported
	6	0	Either Low-value payment or CTTA checking not supported
	5	1	New card checking supported
	4	0	PIN Try Limit Exceeded checking supported
	3	0	Non-currency matching offline transaction not supported

Byte	Bit	Value	Definition
	2	0	The card does not prefer to choose contactIC card online transaction first
	1	0	Do not return offline purchase amount available.
2	8	0	RFU
	7	0	Non-currency-matching transaction not allowed
	6	1	If it is a new card and card reader only supports offline transactions, not decline the transaction.
	5	0	Card Transaction Log
	4-1	00000	RFU
3	8	0	Currency-matching transaction not support online PIN
	7	0	Non-currency-matching transaction not support online PIN
	6	0	For non-currency-matching transactions, CVM not required
	5	1	Signature supported
	4-1	0000	RFU
4	8-7	00	RFU
4	6	1	ODA for online authorization supported
4	5-1	00000	RFU

5.7.1.5 Card Transaction Qualifiers

'00 00' hexadecimal

Byte	Bit	Value	Definition
1	8	0	Online PIN not required
1	7	0	Signature not required
1	6	1	If offline data authentication fails and the terminal is capable of online processing, for online processing
1	5	0	If offline data authentication fails and the terminal supports contact IC cards, not terminate the transaction.
1	4-1	0000	RFU
2	8	0	CDCVM NOT performed
2	7-1	0000000	RFU

5.7.1.6 Issuer Application Data

Byte	Bit	Hexadecimal Initial Value	Conditions
1	8-1	07	Length
2	8-1	Issuers fill in according to actual situation	Derivation key index
3	8-1	01	Cryptogram version No.
4-7	8-1	03 00 00 00	Card verification result (CVR)
8	8-1	01	Algorithm identification
9	8-1	0A	If issuer self-defined data exist, self-defined data length (15 max)
10	8-1	01	IDD ID
11-24	8-1		Issuer defined Data

5.7.1.7 Card Additional Function Indicator

Byte	Bit	Hexadecimal Initial Value	Definition
1	8	0	Does not support extended application record data R-MAC protection
	7	1	ODA for online authorization supported
	6	0	RFU
	5	0	Does not support credit-mode low-value payment
	4	0	Does not support dormant card management
	3	0	Does not support bilateral security authentication
	2	0	Does not support fee deduction by segment, does not support offline pre-authorization
	1	0	Does not support fee deduction by segment

Appendix A

A.1 UICS Data Definition Supplemental Description

A.1.1 CVM Table

In the debit/credit application personalization templates, the recommended CVM table values already cover the majority of debit card and credit card basic regulations and common cardholder usage habits. But since the risk strategies and the target population of each issuer is not identical, during issuer card issuance, adjustments can be made based on actual circumstances to the CVM table to meet specific needs.

A.1.2 Record Data Used in Static Signatures

Data records used in static signatures shall be labeled with the 4th byte of the AFL, shall not contain data that can be updated through issuer script, and is recommended to be composed of the following data:

Data Object Name	Tag	Length
Application expiration date	5F24	3
Application expiration date	5F25	3
Application primary account number	5A	6
Application usage control	9F07	2
Application version number	9F08	2
Issuer country code	5F28	2
Cardholder Verification Method (CVM) List	8E	Var.
IAC - Reject	9F0D	5
IAC - Online	9F0E	5
IAC - Default	9F0F	5

Data Object Name	Tag	Length
CDOL1	8C	Var.
CDOL2	8D	Var.

If card supports contactless qUICS, then the qUICS signature static application data shall not include application primary account serial number (5F34). The reason is: IF 5F34 is included in the signature's static application data, then this data may be repeatedly read by terminal during transactions, resulting in abnormal transaction.

A.1.3 AFL of qUICS

If card supports contactless fast payment applications (qUICS), then recommend making card authentication related data (tag "9F69") as the last record in the qUICS application AFL list, and that the last record only include this data element or the last record only includes 9F74.

The reason being: In some circumstances, after the card sends out all records, the terminal can still see offline data authentication failure due to not completely receiving the last record. Making the last record of the AFL list a short record can lower the probability of the terminal not being able to completely read the last record due to data being too long.

A.1.4 Card Validity Period

The expiration date in the back of card magnetic stripe, the application expiration date for the prime debit/credit application in the chip, and the expiration date in the magnetic track 2 equivalent data must remain consistent.

Also, if the front of the card is printed with the effective date, then the front of card effective date must also remain consistent with the application expiration date of the primary debit/credit application in the chip.

Issuer shall ensure the effective date of the certification authority public key is longer than the effective date printed on the card surface.

A.1.5 “ans” Data Format Description

Except for application preferred name (Tag “9F12”), all data object with data format of “ans” shall use single byte codec numeric, alphabet and special characters, etc. Please refer to EMV 4.3 Book 4, appendix B for details. Application preferred name may use special codec format. If terminal support application preferred name codec, then it shall use application preferred name codec format to display or print characters, or else display or print application label.

A.2 UICS Personalization Template Usage Guide

This UICS specification lists 17 templates, including standard debit/credit application personalization templates (templates 1 thru 9) and low-value payment and qUICS functionality personalization templates (templates 11 thru 17).

Standard debit/credit application personalization templates defined the personalization data values for debit IC cards, credit IC cards, quasi-credit IC cards, and social security cards loaded with financial functionalities (financial applications); low-value payment and qUICS functionality personalization templates define the personalization data values for electronic cash and qUICS process, each template including Low-value Payment based on debit/credit (contact) and Low-value Payment based on fast debit/credit (contactless type) personalization data. During issuer usage, template could be chosen based on card type, as shown in figures below:

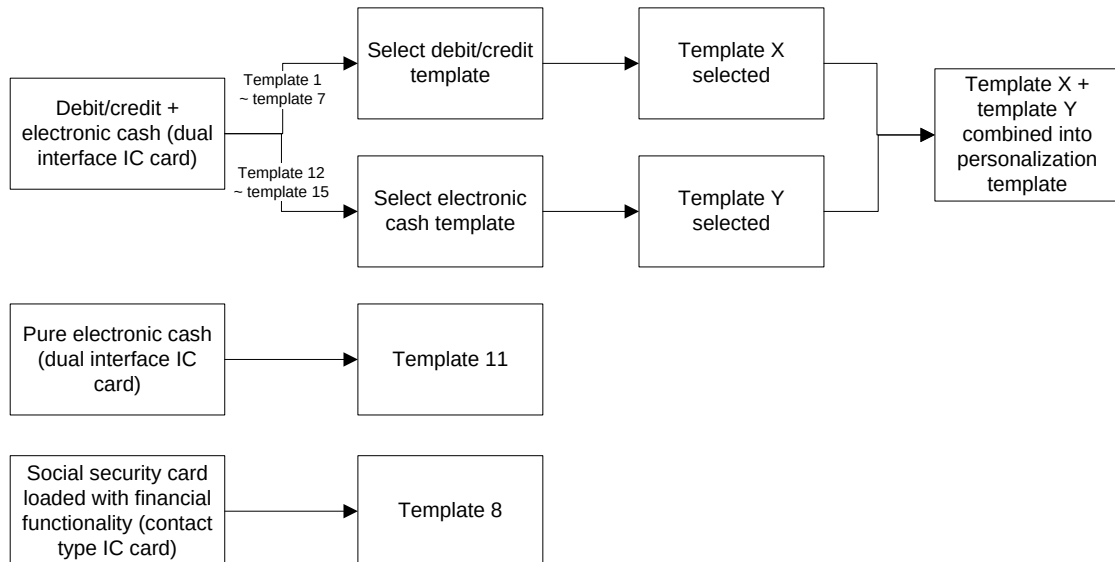


Figure 1: Diagram of personalization template selection

For example, an issuer that issues a "debit/credit + electronic cash" IC card can first choose a debit or credit card (including quasi-credit) template from the standard debit/credit application personalization templates, then choose a low-value payment and qUICS template from the electronic cash functionality personalization templates, combining them into a personalization template for personalizing IC card.

For an issuer that issues "social security card loaded with financial functionality" IC card, the debit application in the social security card can use template 8 for IC card personalization.

For issuer that issues "pure electronic cash card" IC card, template 11 can be selected for IC card personalization.

A.2.1 Debit Card Template Selection

This template provides 4 debit card templates, namely template 1, template 2, template 8 and template 9. The CVMs for these 4 templates only supports online PIN (no CVM excluded), with the difference being, template 1 permits BYPASS PIN, template 2 and template 8 do not permit BYPASS PIN. Template 9 is the template for signature debit card, which gives priority to use signature for cardholder verification in face-to-face non-cash transaction.

In the template 1 setup, the cardholder fails to enter PIN for any reason, so long as the cardholder successfully authenticates in the end (such as execution NO CVM processing), then it is permitted for transaction to request online, the terminal shall not reject transaction. This template is relatively permissive on cardholder verification conditions, and issuer can select this template based on business need.

In the template 2 and template 8 setup, the cardholder fails for any reason to enter PIN, regardless of whether the cardholder successfully authenticates in the end, transaction will be rejected. These templates are relatively strict on cardholder verification conditions, and issuer can select these templates based on business need.

Template 8 can also be used in the personalization of financial applications in financial social security cards, transactions involving products, services, and cash setup under this template is only permitted domestically, for above type transaction that occur abroad terminal will reject.

A.2.2 Credit Card Template Selection

This template provides 4 credit card templates, namely template 3, template 4, template 5, and template 6. These 4 templates differ on CVMs and related IAC setting.

Template 3 is relatively permissive overall, supports online PIN. When online PIN is not entered for any reason or if the terminal does not support online PIN, then cardholder authentication could be completed through signature. So long as cardholder authentication succeeds, then terminal would not reject transaction.

Template 4 is relatively permissive overall, supports signature. When terminal does not Signature supported, cardholder authentication can be completed through online PIN. So long as cardholder authentication succeeds, then terminal would not reject transaction.

Template 5 is relatively strict overall, supports online PIN. Online PIN execution failure results in transaction rejection; if terminal does not support online PIN, then cardholder authentication could be completed through signature. If cardholder authentication fails, then terminal rejects transaction.

Template 6 is relatively permissive overall, supports offline PIN. If terminal does not support offline PIN or cardholder fails to enter offline PIN, then cardholder

authentication could be completed through signature. So long as cardholder authentication succeeds, then terminal would not reject transaction.

Template 7 is the quasi-credit card template, setting are relatively permissive, supports online PIN. When online PIN execution fails, then cardholder authentication could be completed through isomorphic signature.

A.2.3 Electronic Cash Template Selection

This template provides 7 electronic cash templates, namely template 11 ~ 17, each template includes Low-value Payment data set (contact electronic cash) based on debit/credit and Low-value Payment (contactless Low-value Payment) data set based on fast debit/credit. Low-value Payment based on debit/credit all support CDA verification method, Low-value Payment based on fast debit/credit all support version 01 fDDA dynamic signatures. Besides template 11, since other templates are bundled with debit applications or credit applications, the contact electronic cash in these templates all Signature supported; because a pure electronic cash template has no bundled debit application or credit application, therefore the contact electronic cash in pure electronic cash template do not Signature supported, and only support NO CVM.

Template 11 is the pure electronic cash template, including standard debit/credit data set, data set for Low-value Payment based on debit/credit, and data set based on fast debit/credit, and therefore can be used independently. Template 12 ~ template 15 only define Low-value Payment type data set, and do not include standard debit/credit data set, therefore when issuer issues "(quasi-)debit/credit + electronic cash" type IC cards, a standard debit/credit application data template can be selected for use from template 1 ~ template 7. Data not included in electronic cash template could be directly referenced from standard debit/credit templates.

Template 12 and template 13 are both common electronic cash templates, with the difference being, template 12 supports contactless transaction log, template 13 does not support contactless transaction log;

Template 14 is a template with electronic cash that support dual currency types, appropriate for Hong Kong-Macao issuers who issue dual currency electronic cash;

Template 15 is a template that supports contactless Low-value Payment extended application electronic cash, appropriate for issuers launching contactless IC card fee deduction by segment or by time Low-value Payment business domestically.

A.3 Country/Region and Currency Code

Issuer can refer to the following table for Issuer Country/Region Code and Application Currency Code.

Country/Region Name	Issuer Country/Region Code (tag '5F28')	Application Currency Code (tag '9F51')
Afghanistan	004	971
Albania	008	008
Algeria	012	012
American Samoa	016	840
Andorra	020	978
Angola	024	973
Anguilla	660	951
Antigua and Barbuda	028	951
Argentina	032	032
Armenia	051	051
Aruba	533	533
Australia	036	036
Austria	040	978
Azerbaijan	031	944
Bahamas	044	044
Bahrain	048	048
Bangladesh	050	050
Barbados	052	052
Belarus	112	933
		974
Belgium	056	978
Belize	084	084
Benin	204	952
Bermuda	060	060
Bhutan	064	356
		064
Bolivia	068	068
		984
Bosnia and Herzegovina	070	977
Botswana	072	072
Bouvet Island	074	578
Brazil	076	986
British Indian Ocean Territory	086	840
Brunei Darussalam	096	096
Bulgaria	100	975
Burkina Faso	854	952
Burundi	108	108
Cambodia	116	116
Cameroon	120	950
Canada	124	124
Cape Verde	132	132
Cayman Islands	136	136
Central African Republic	140	950
Chad	148	950

Country/Region Name	Issuer Country/Region Code (tag '5F28')	Application Currency Code (tag '9F51')
Chile	152	152
		990
China	156	156
Christmas Island	162	036
Cocos (Keeling) Islands	166	036
Colombia	170	170
		970
Comoros	174	174
Congo	178	950
Congo (the Democratic Republic of the)	180	976
Cook Islands	184	554
Costa Rica	188	188
Côte D'Ivoire	384	952
Croatia	191	191
Cuba	192	192
		931
Curaçao	531	532
Cyprus	196	978
Czech Republic	203	203
Denmark	208	208
Djibouti	262	262
Dominica	212	951
Dominican Republic	214	214
East Timor	626	360
Ecuador	218	840
Egypt	818	818
El Salvador	222	222
		840
Equatorial Guinea	226	950
Eritrea	232	232
Estonia	233	978
Ethiopia	230	230
Faroe Islands	234	208
Falkland Islands (Malvinas)	238	238
Fiji	242	242
Finland	246	978
France	250	978
France, Metropolitan	249	978
French Guiana	254	978
French Polynesia	258	953
French Southern Territories	260	978
Gabon	266	950
Gambia	270	270
Georgia	268	981

Country/Region Name	Issuer Country/Region Code (tag '5F28')	Application Currency Code (tag '9F51')
Germany	276	978
Ghana	288	936
Gibraltar	292	292
Greece	300	978
Greenland	304	208
Grenada	308	951
Guadeloupe	312	978
Guam	316	840
Guatemala	320	320
Guinea	324	324
Guinea-Bissau	624	952
Guyana	328	328
Haiti	332	332
		840
Heard and McDonald Islands	334	036
Honduras	340	340
Hong Kong	344	344
Hungary	348	348
Iceland	352	352
India	356	356
Indonesia	360	360
Iran, Islamic Republic of	364	364
Iraq	368	368
Ireland	372	978
Isle of Man	833	826
Israel	376	376
Italy	380	978
Jamaica	388	388
Japan	392	392
Jordan	400	400
Kazakhstan	398	398
Kenya	404	404
Kiribati	296	036
Korea, Democratic People's Republic of	408	408
Korea, Republic of	410	410
Kuwait	414	414
Kyrgyzstan	417	417
Lao People's Democratic Republic	418	418
Latvia	428	978
Lebanon	422	422
Lesotho	426	426
		710
Liberia	430	430

Country/Region Name	Issuer Country/Region Code (tag '5F28')	Application Currency Code (tag '9F51')
Libya	434	434
Liechtenstein	438	756
Lithuania	440	978
Luxembourg	442	978
Macau	446	446
Macedonia	807	807
Madagascar	450	969
Malawi	454	454
Malaysia	458	458
Maldives	462	462
Mali	466	952
Malta	470	978
Marshall Islands	584	840
Martinique	474	978
Mauritania	478	478
Mauritius	480	480
Mayotte	175	978
Mexico	484	484
Micronesia	583	840
Midway Islands	488	840
Moldova, Republic of	498	498
Monaco	492	978
Mongolia	496	496
Montenegro	499	978
Montserrat	500	951
Morocco	504	504
Mozambique	508	943
Myanmar	104	104
Namibia	516	516
Nauru	520	710
Nauru	520	036
Nepal	524	524
Netherlands	528	978
Netherlands Antilles	530	532
New Caledonia	540	953
New Zealand	554	554
Nicaragua	558	558
Niger	562	952
Nigeria	566	566
Niue	570	554
Norfolk Island	574	036
Northern Mariana Islands	580	840
Norway	578	578
Oman	512	512

Country/Region Name	Issuer Country/Region Code (tag '5F28')	Application Currency Code (tag '9F51')
Pakistan	586	586
Palau	585	840
Palestinian (Palestine)	275	840
Panama	591	590
		840
Papua New Guinea	598	598
Paraguay	600	600
Peru	604	604
Philippines	608	608
Pitcairn	612	554
Poland	616	985
Portugal	620	978
Puerto Rico	630	840
Qatar	634	634
Réunion	638	978
Romania	642	946
Russian Federation	643	643
Rwanda	646	646
Saint Barthelemy	652	978
Saint Martin (French part)	663	978
Samoa	882	882
San Marino	674	978
Sao Tome and Principe	678	678
Saudi Arabia	682	682
Senegal	686	952
Serbia and Montenegro	891	941
Serbia	688	941
Seychelles	690	690
Sierra Leone	694	694
Singapore	702	702
Sint Maarten (Dutch part)	534	532
Slovakia	703	978
Slovenia	705	978
Solomon Islands	090	090
Somalia	706	706
South Africa	710	710
South Georgia and The South Sandwich Islands	239	
South Sudan	728	728
Spain	724	978
Sri Lanka	144	144
St. Helena	654	654
St. Kitts-Nevis	659	951
St. Lucia	662	951
St. Pierre and Miquelon	666	978

Country/Region Name	Issuer Country/Region Code (tag '5F28')	Application Currency Code (tag '9F51')
Saint Vincent and the Grenadines	670	951
Sudan	736	938
Suriname	740	968
Svalbard and Jan Mayen Islands	744	578
Swaziland	748	748
Sweden	752	752
Switzerland	756	756
Syrian Arab Republic	760	760
Taiwan	158	901
Tajikistan	762	972
Tanzania, United Republic of	834	834
Thailand	764	764
Togo	768	952
Tokelau	772	554
Tonga	776	776
Trinidad and Tobago	780	780
Tunisia	788	788
Turkey	792	949
Turkmenistan	795	934
Turks and Caicos Islands	796	840
Tuvalu	798	036
U.S. Minor Outlying Islands	581	840
Uganda	800	800
Ukraine	804	980
United Arab Emirates	784	784
United Kingdom	826	826
United States	840	840
Uruguay	858	858
Uzbekistan	860	860
Vanuatu	548	548
Holy See (Vatican City State)	336	978
Venezuela	862	937
Vietnam	704	704
Virgin Islands, British	092	840
Virgin Islands, U.S.	850	840
Wake Island	872	840
Wallis and Futuna Islands	876	953
Western Sahara	732	504
Yemen	887	886
Zambia	894	967
Zimbabwe	716	932